

Image Recognition



Computer Vision

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Recognition and Learning

- Image Features and Categorization
- Classifiers
- Neural Networks
- Convolutional Neural Networks
- Object Detection
- Segmentation
- Image Generation

What do you see in this image?

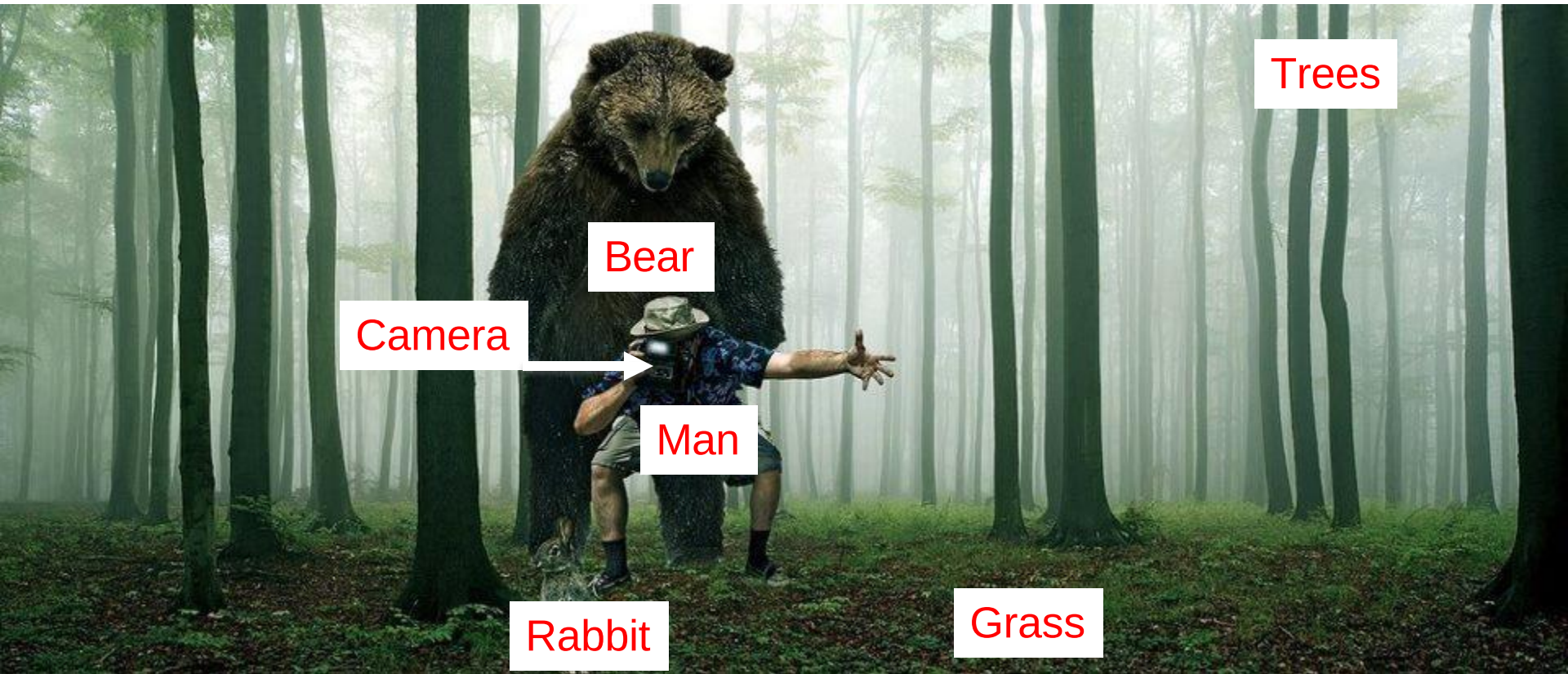


What do you see in this image?



Forest

What do you see in this image?



Forest

Describe, predict, or interact with the object based on visual cues



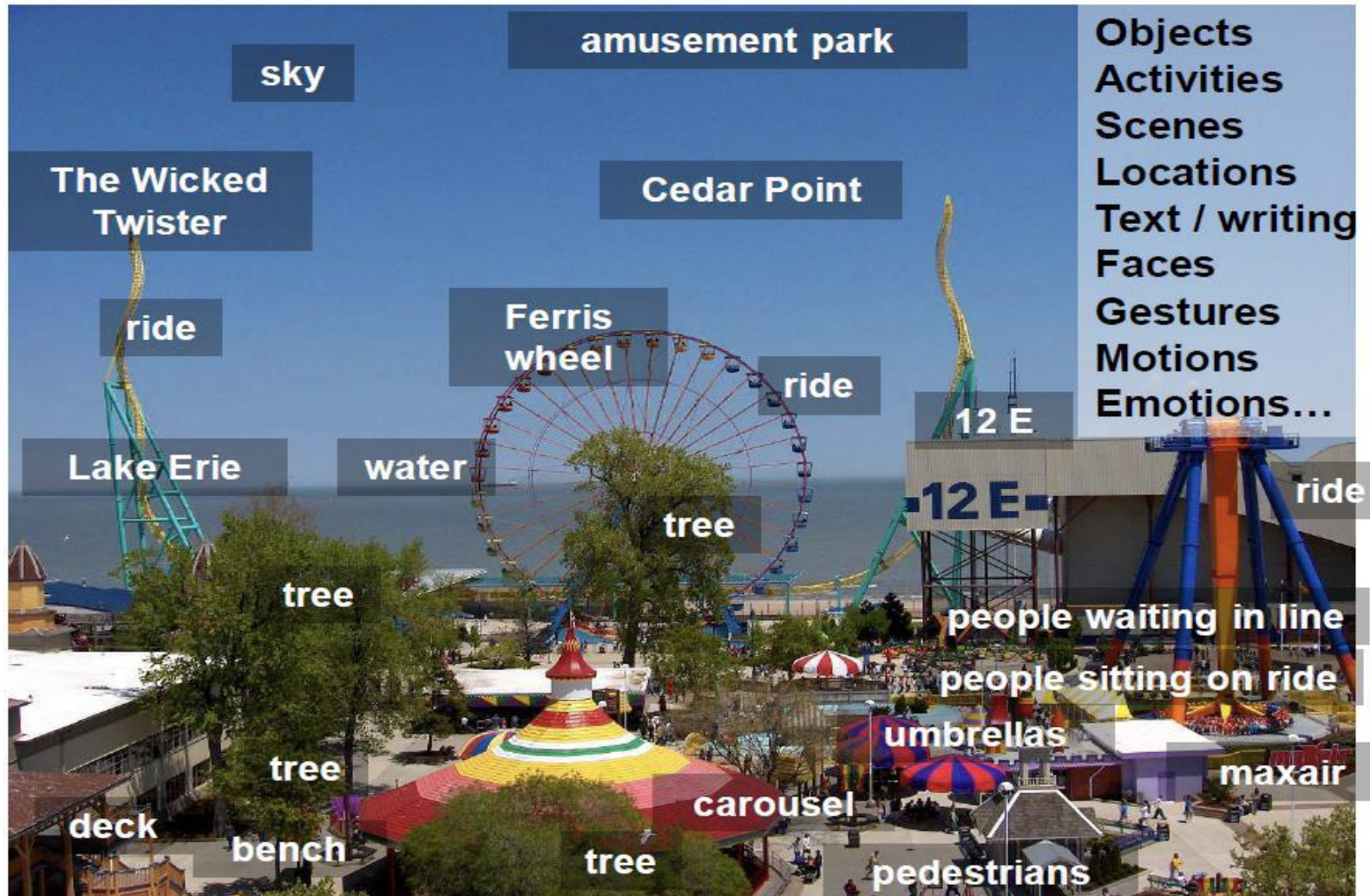
Is it **dangerous**?

Is it **alive**?

How **fast** does it run?

Does it have a **tail**?

Image Recognition



Semantic
information
Metric 3D
information

Image Recognition:

- The ability of a system to detect, interpret, and understand objects or patterns in an image

Ex:

- Image Classification, Multi-label Classification
- Object detection
- Semantic/instance Segmentation
- Pose detection
- Anomaly detection
- Image captioning

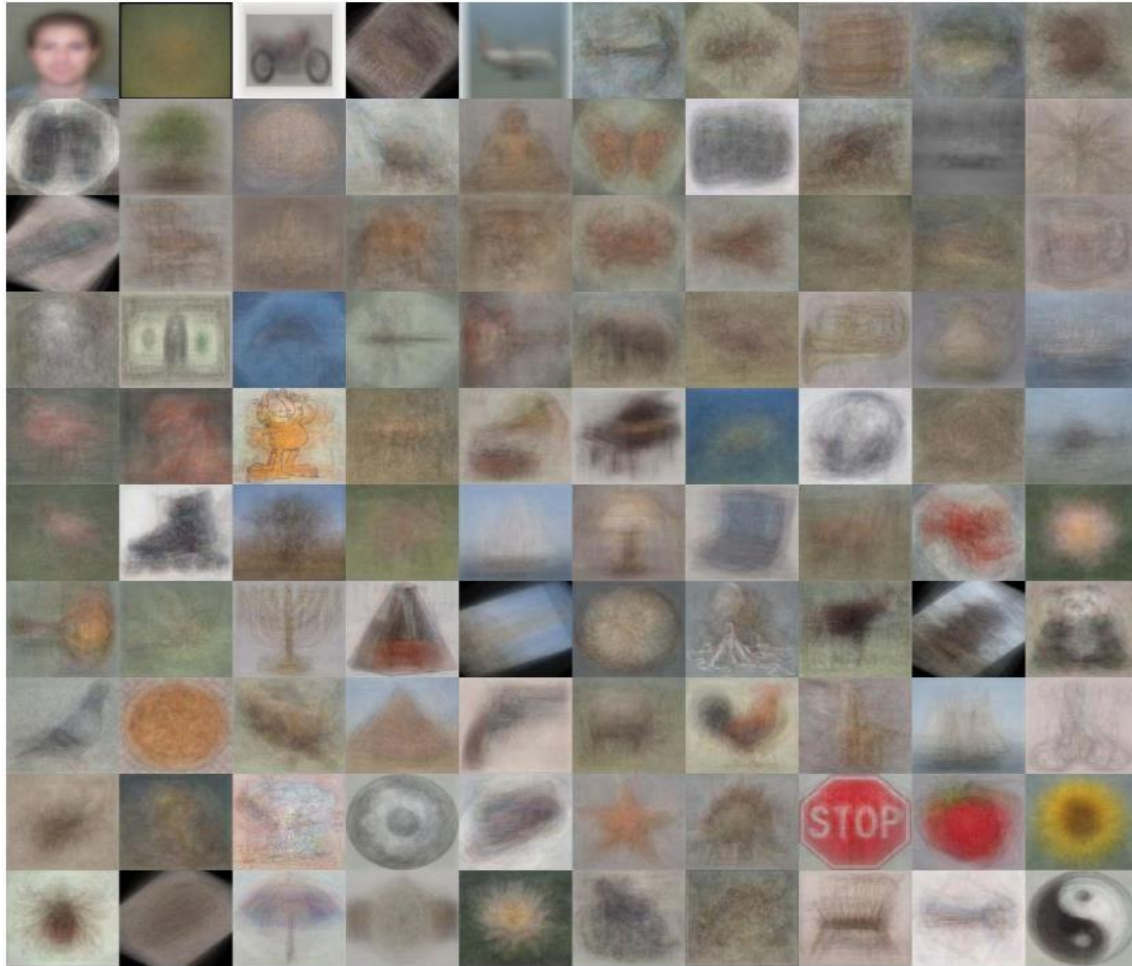
Image categorization

- Cat vs Dog



Image categorization

- Object recognition



Caltech 101 Average Object Images

Why do we care about categories?

- From an object's category, we can make predictions about its behavior in the future, beyond of what is immediately perceived.
- Pointers to knowledge
 - Help to understand individual cases not previously encountered

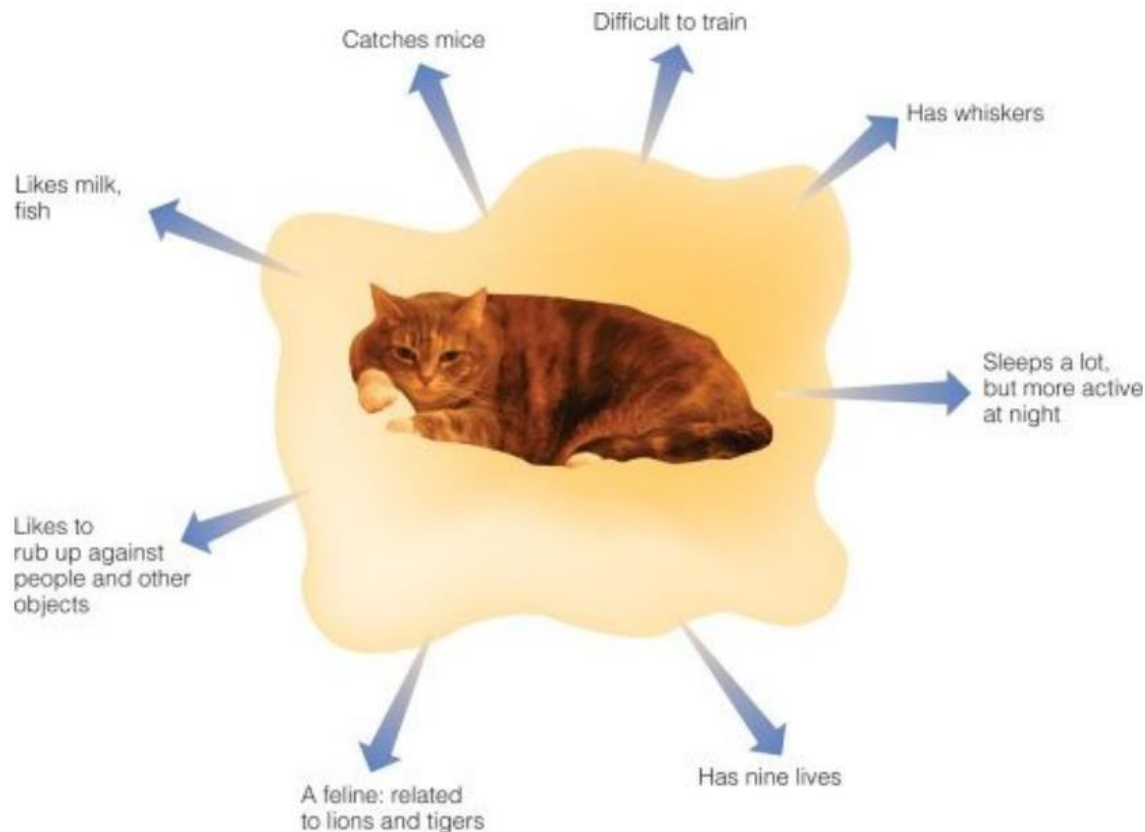


Image categorization

- Place recognition



Places Database [[Zhou et al. NIPS 2014](#)]

Image categorization

- Image style recognition



HDR



Macro



Baroque



Rococo



Vintage



Noir



Northern Renaissance



Cubism



Minimal



Hazy



Impressionism



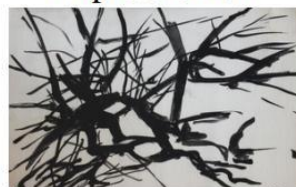
Post-Impressionism



Long Exposure



Romantic



Abs. Expressionism



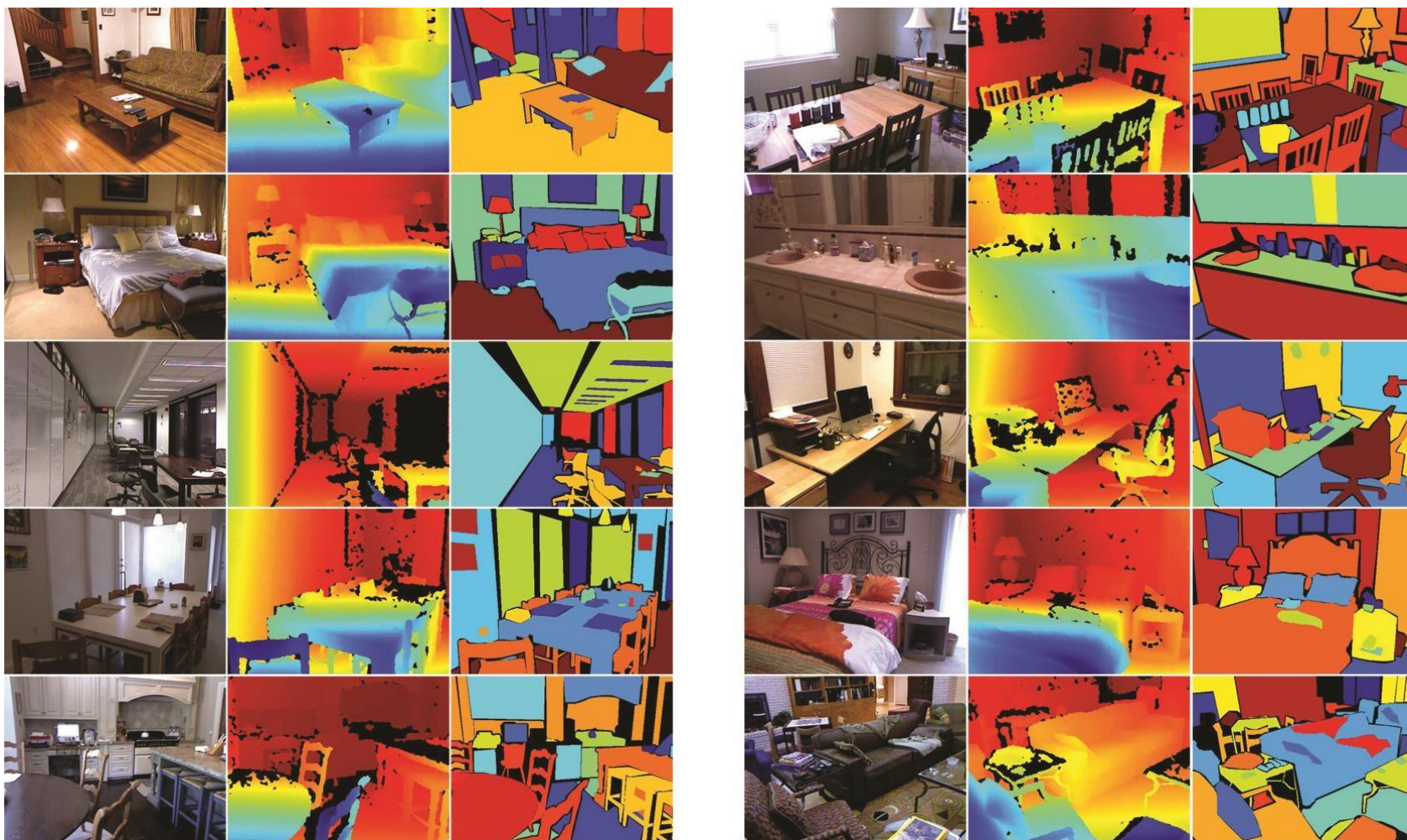
Color Field Painting

Flickr Style: 80K images covering 20 styles.

Wikipaintings: 85K images for 25 art genres.

Region categorization

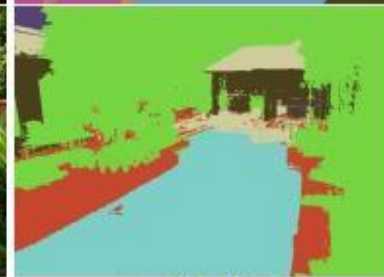
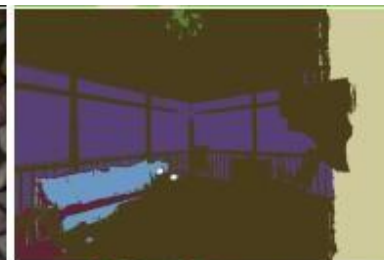
- Semantic segmentation from RGBD images



[[Silberman et al. ECCV 2012](#)]

Region categorization

- Material recognition

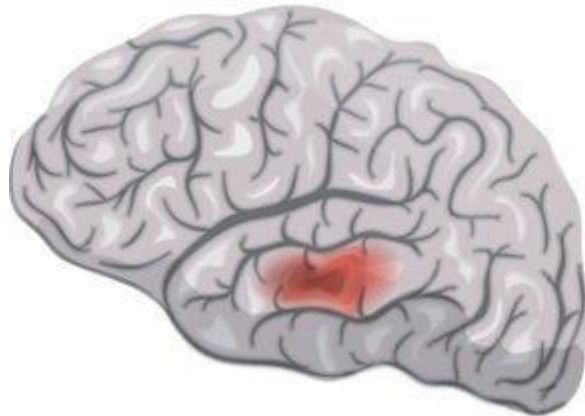


[[Bell et al. CVPR 2015](#)]

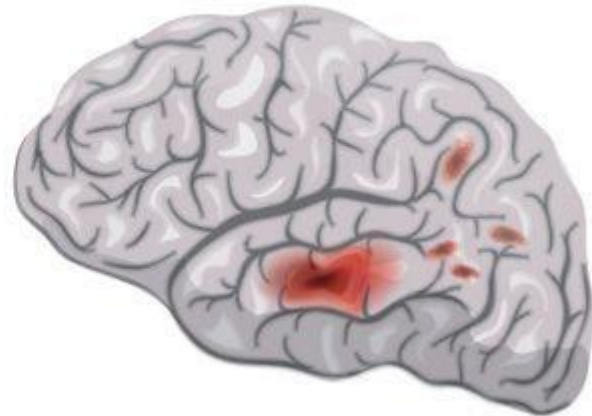
Image categorization

- Primary Tumor vs Metastasis

Brain Cancer



the primary tumor



metastasis

Recognition: "Simple" patterns



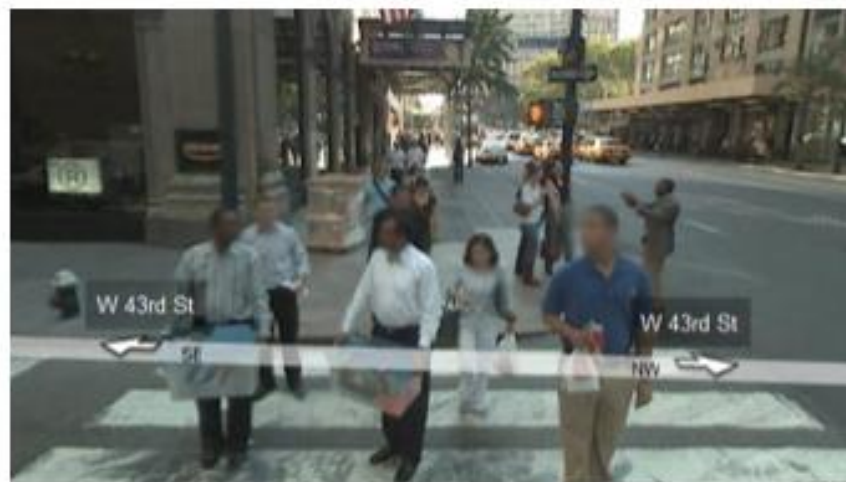
4YCH428

4YCH428

4YCH428

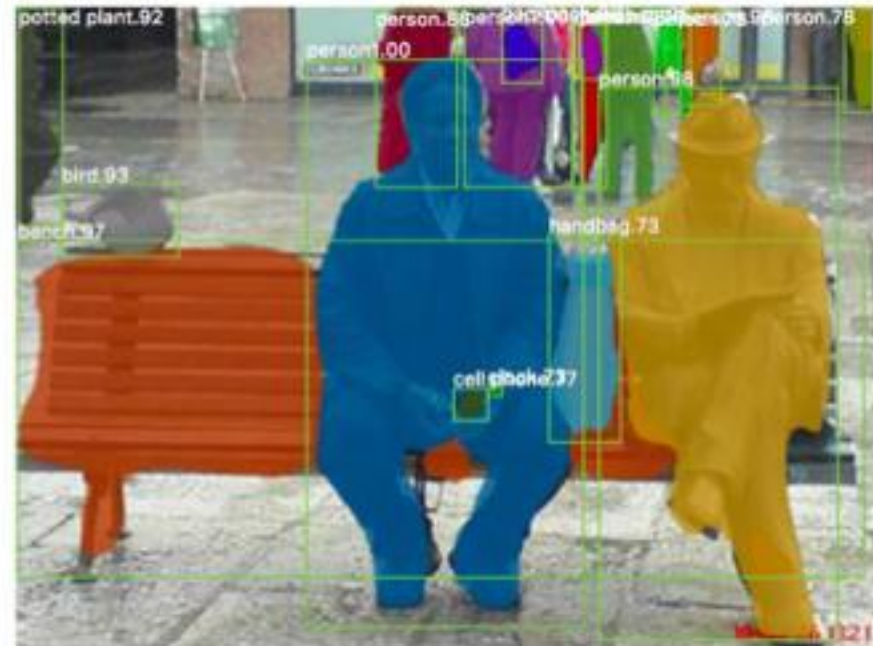


Recognition: Faces



Slide adapted from SVETLANA LAZEBNIK

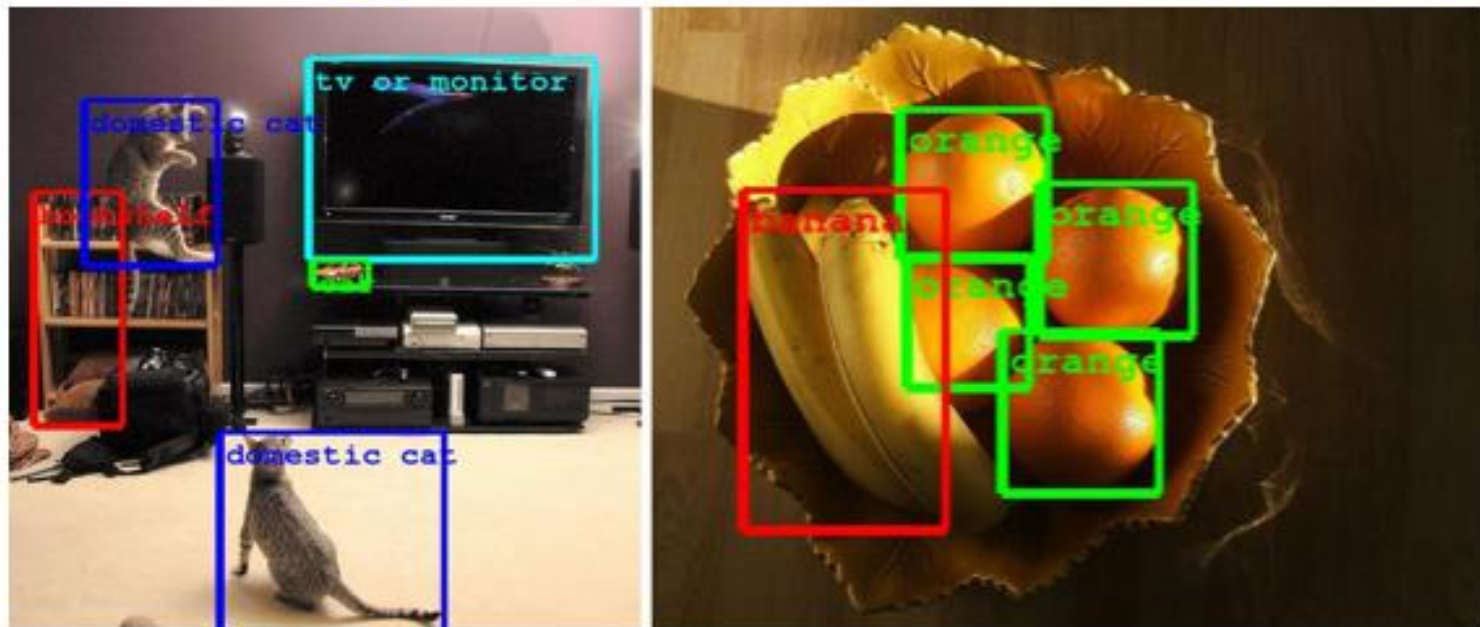
Object detection, instance segmentation



K. He, G. Gkioxari, P. Dollar, and R. Girshick, [Mask R-CNN](#),
ICCV 2017 (Best Paper Award)

Slide adapted from SVETLANA LAZEBNIK

Recognition: General categories



- [Computer Eyesight Gets a Lot More Accurate](#), NY Times Bits blog, August 18, 2014
- [Building A Deeper Understanding of Images](#), Google Research Blog, September 5, 2014



Applications of CV



Autonomous Vehicles

Credit: smoothgrover22, Flickr CC License



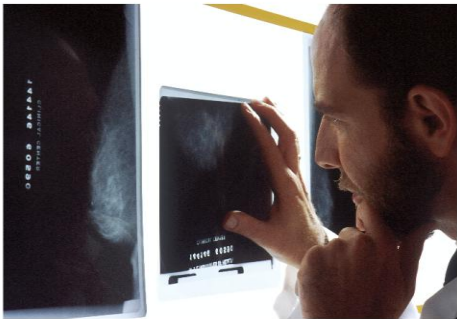
Surveillance

Credit: Yeong Nam, Flickr CC License



Factory Automation

Credit: KUKA Roboter GmbH, Bachmann



Medical Imaging

Credit: National Cancer Institute



Human-Computer Interaction

Credit: Vancouver Film School



Visual Effects

Credit: AntMan3001, Flickr CC License

The statistical learning framework

- Apply a prediction function to a feature representation of the image to get the desired output:

$$f(\text{apple image}) = \text{"apple"}$$

$$f(\text{tomato image}) = \text{"tomato"}$$

$$f(\text{cow image}) = \text{"cow"}$$

The statistical learning framework

$$y = f(x)$$

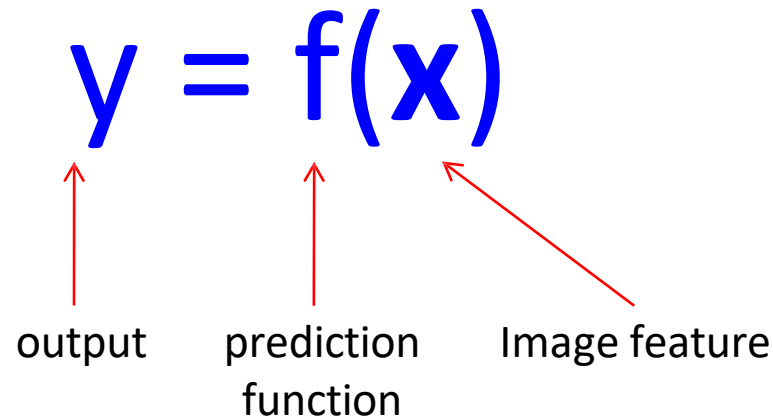
A diagram illustrating the statistical learning framework. The equation $y = f(x)$ is written in large blue font. Below the equation, three labels are positioned: "output" under y , "prediction function" under f , and "Image feature" under x . Red arrows point from each label to its corresponding symbol in the equation: one arrow points from "output" to y , one from "prediction function" to f , and one from "Image feature" to x .

output

prediction
function

Image feature

The statistical learning framework

$$y = f(x)$$


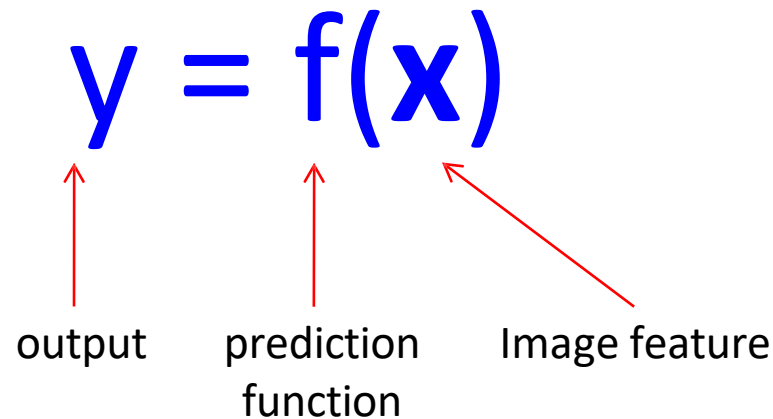
output prediction
 function

Image feature

The diagram shows the equation $y = f(x)$ in blue. Three red arrows point from labels below to the components of the equation: one from 'output' to y , one from 'prediction function' to f , and one from 'Image feature' to x .

- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, estimate the prediction function f by minimizing the prediction error on the training set

The statistical learning framework

$$y = f(x)$$


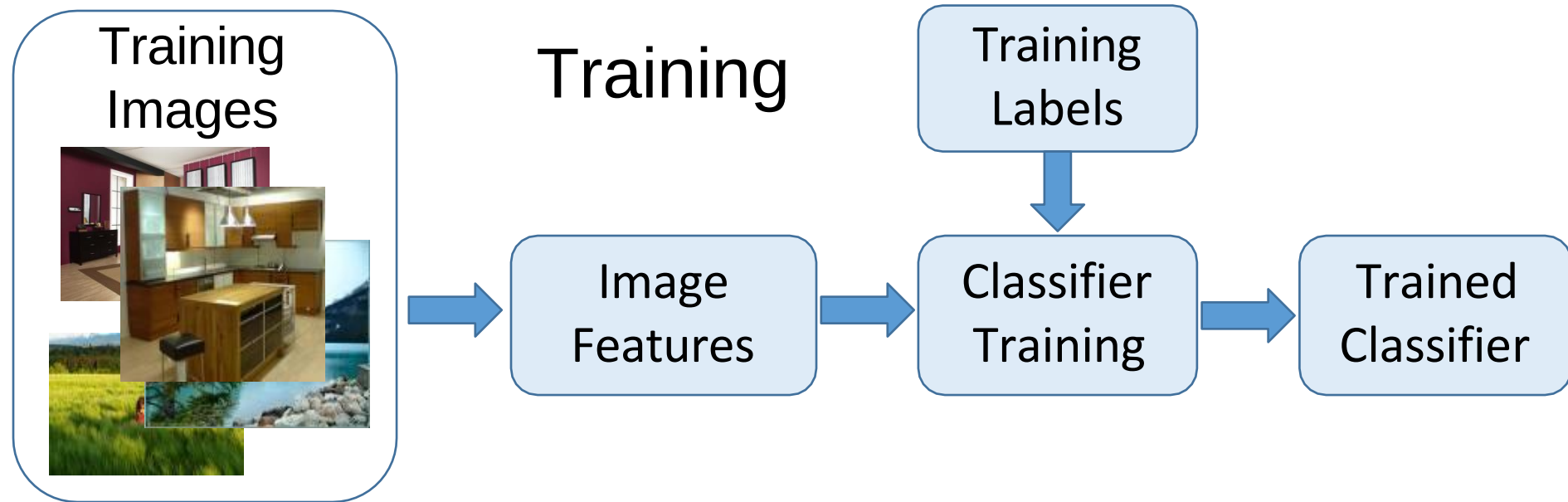
output prediction
 function

Image feature

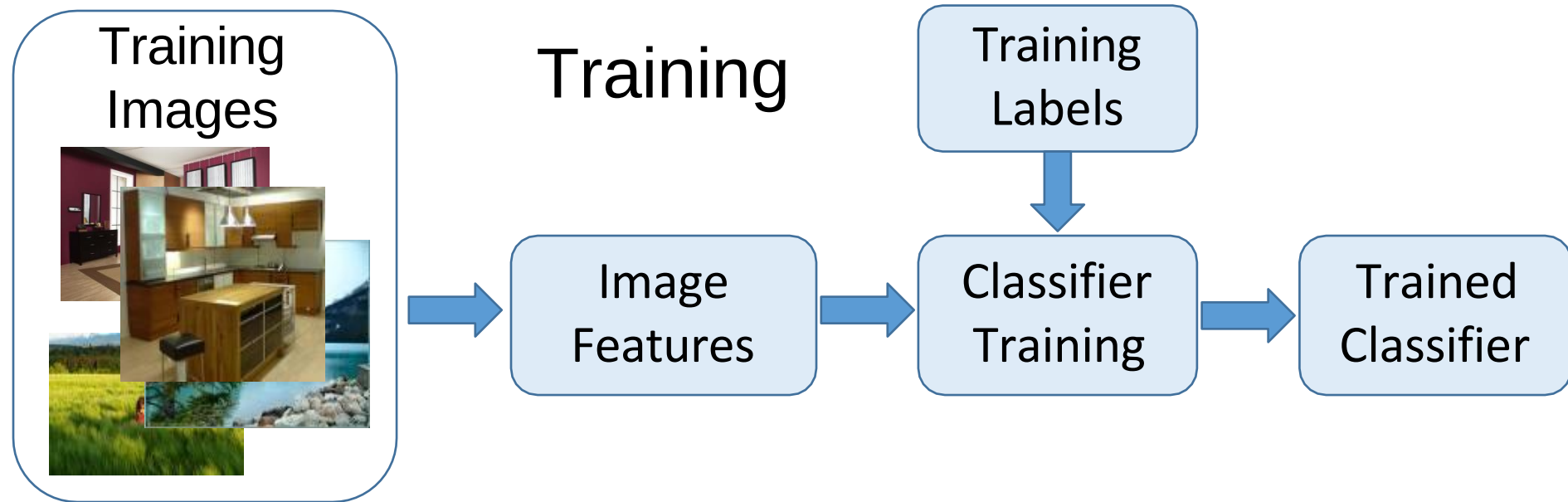
The diagram shows the equation $y = f(x)$ in blue. Below it, three red arrows point upwards to the components of the equation: one to y , one to f , and one to x . Below the arrow pointing to y is the label 'output'. Below the arrow pointing to f is the label 'prediction function'. Below the arrow pointing to x is the label 'Image feature'.

- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, estimate the prediction function f by minimizing the prediction error on the training set
- **Testing:** apply f to a never before seen *test example* $\mathbf{x}$ and output the predicted value $y = f(\mathbf{x})$

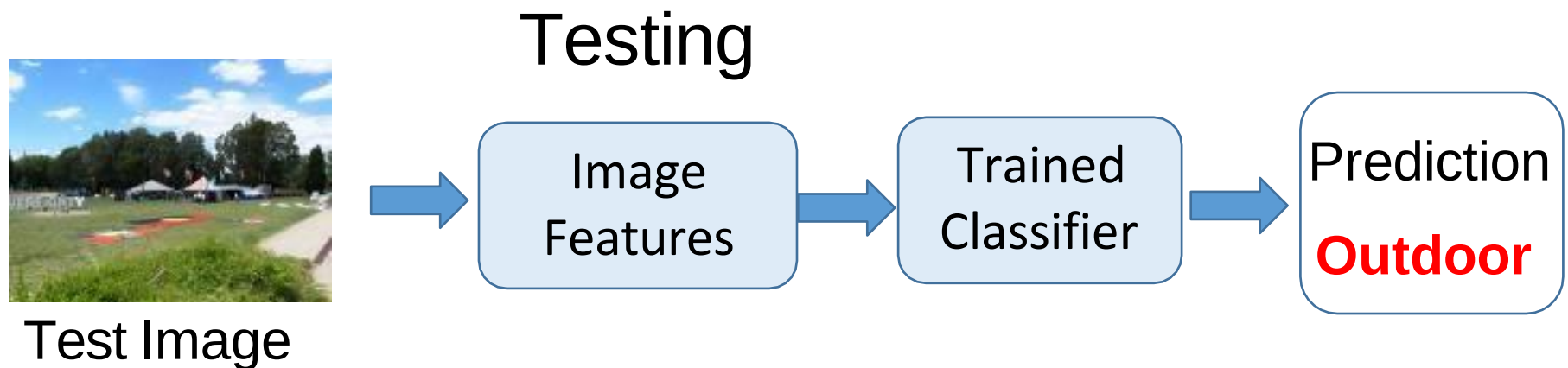
Training phase



Training phase



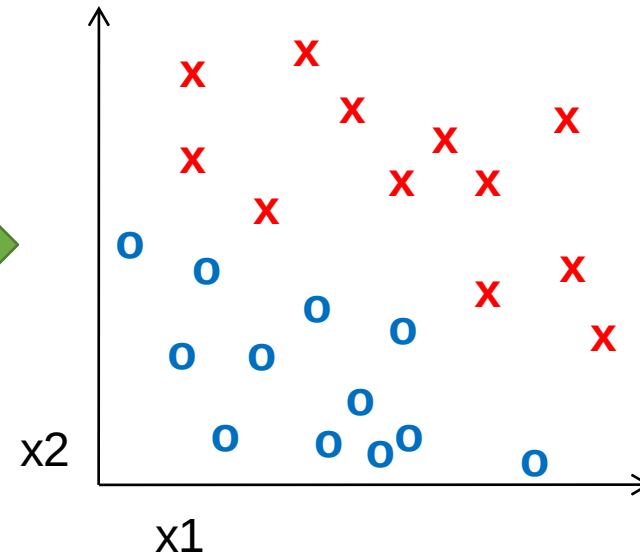
Testing phase



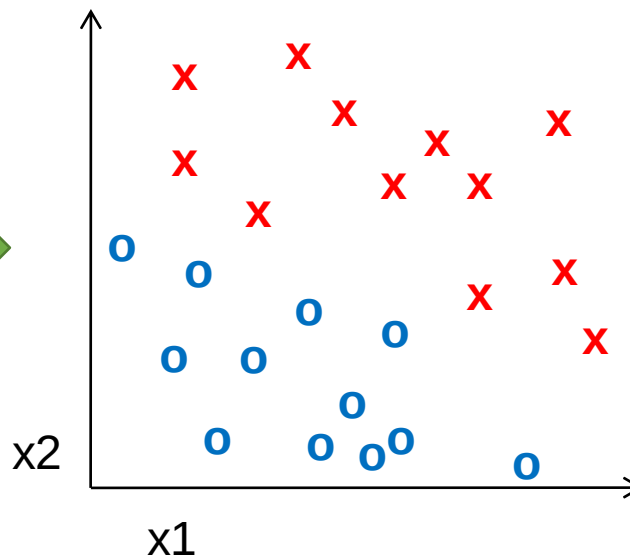
- **Image features:** map images to feature space



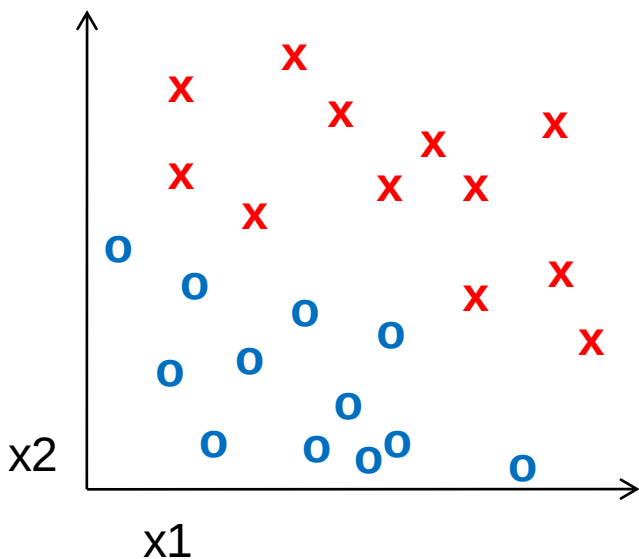
- **Image features:** map images to feature space



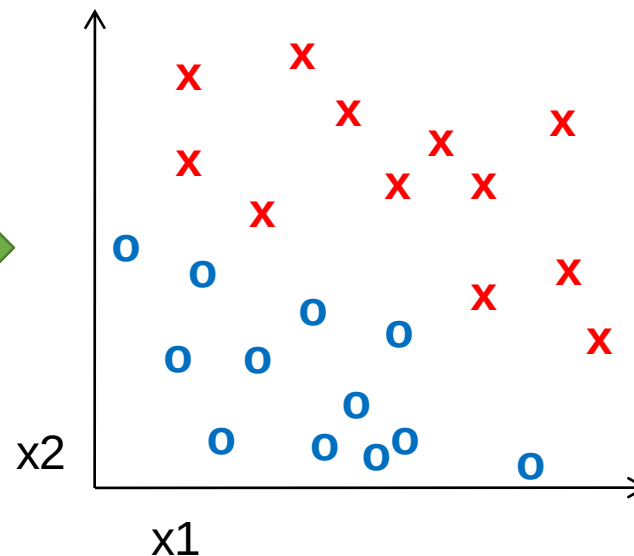
- **Image features:** map images to feature space



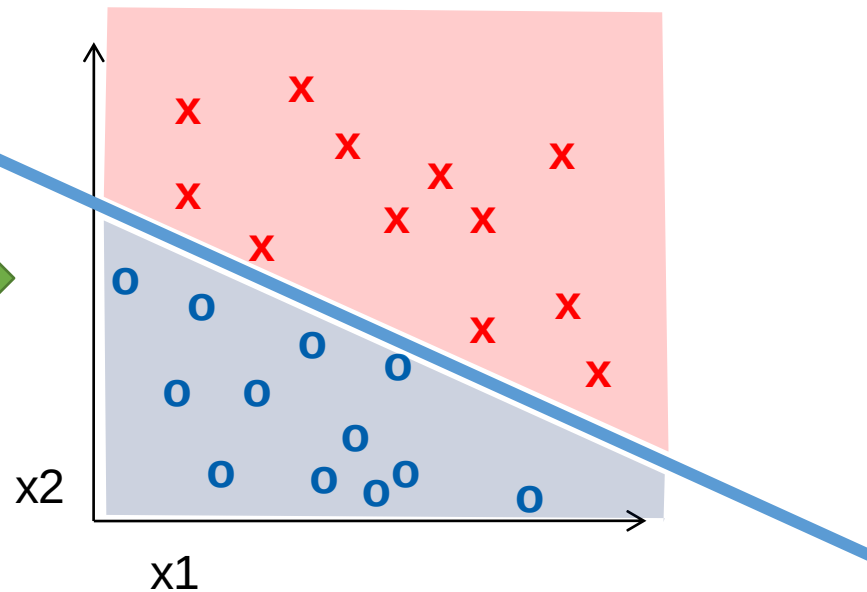
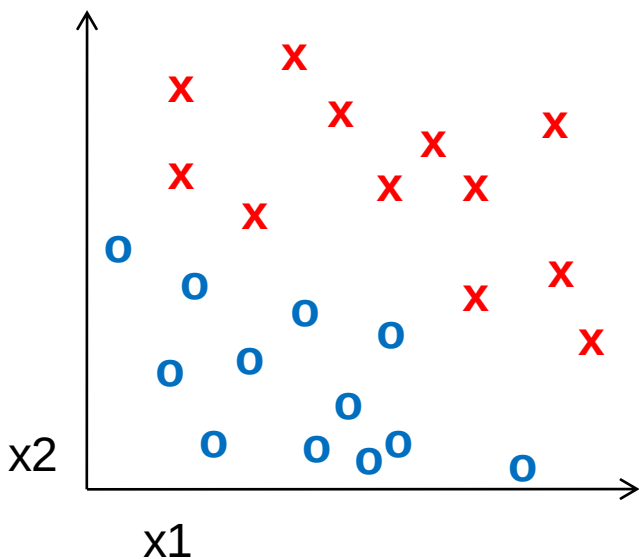
- **Classifiers:** map feature space to label space



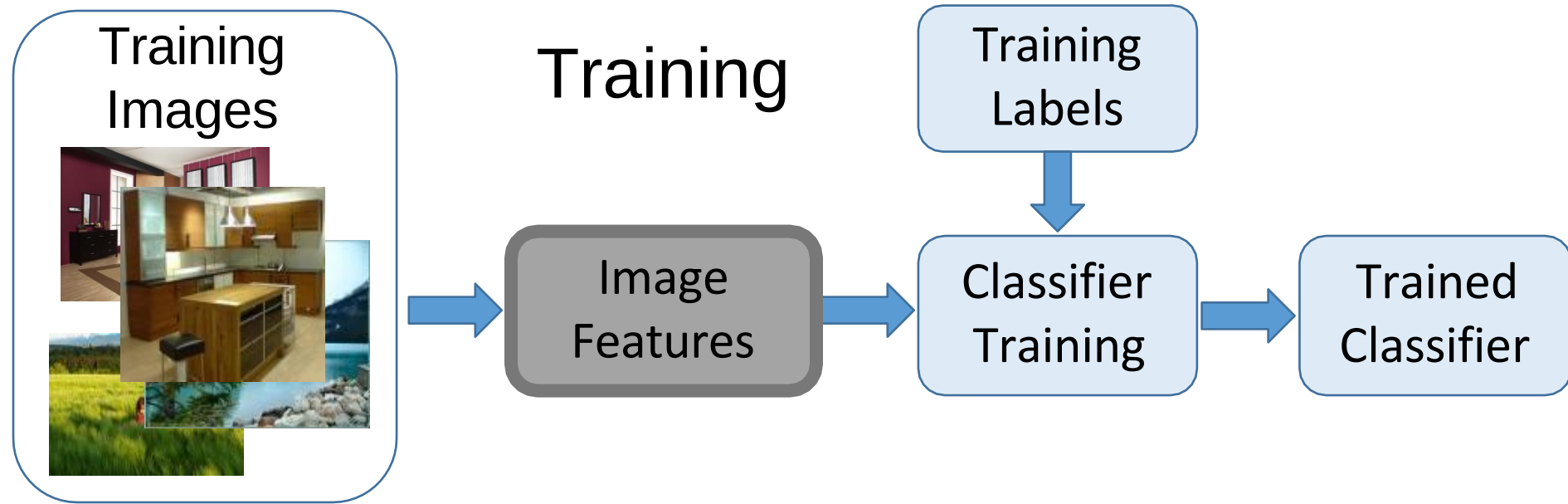
- **Image features:** map images to feature space



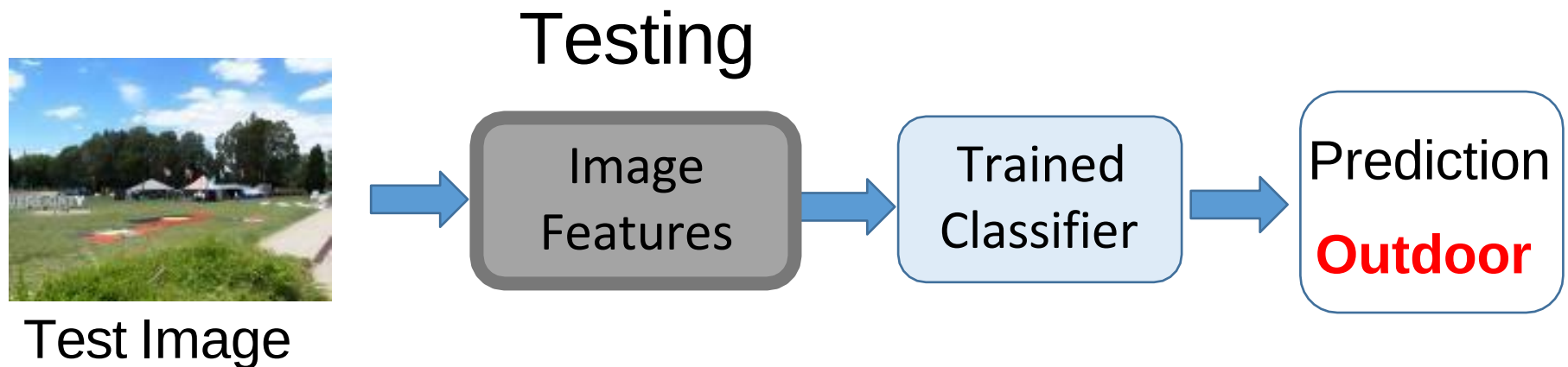
- **Classifiers:** map feature space to label space



Training phase



Testing phase



Q: What are good features for...

- recognizing a beach?



Q: What are good features for...

- recognizing cloth fabric?



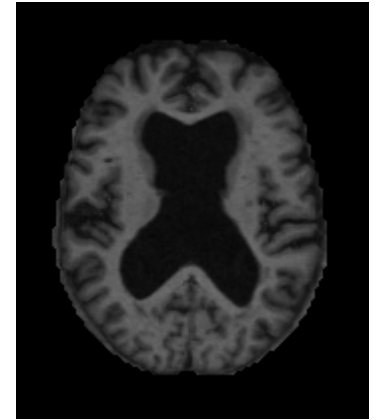
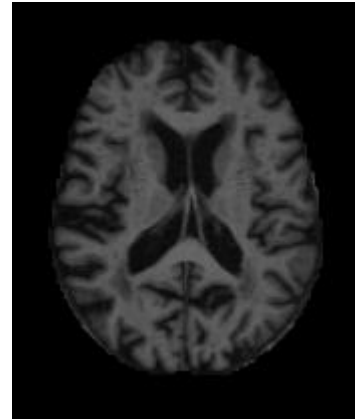
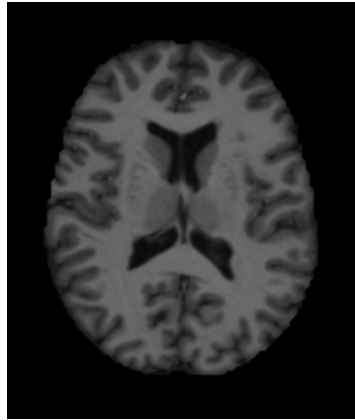
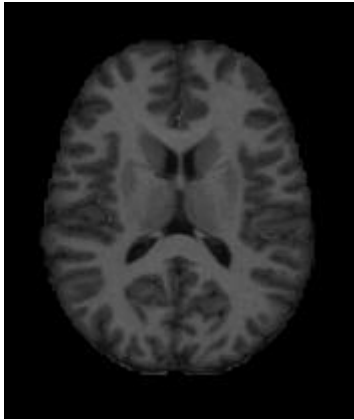
Q: What are good features for...

- recognizing a mug?



Q: What are good features for...

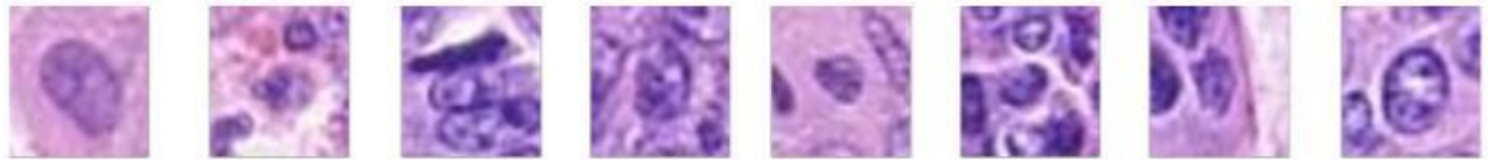
- recognizing the nodule in MRI data?



Q: What are good features for...

- recognizing the type of colon cancer?

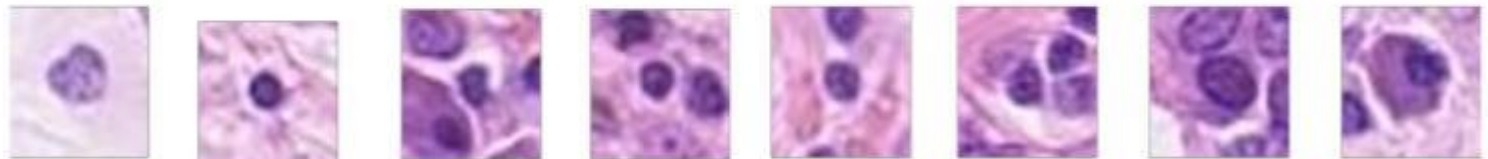
‘Epithelial’



‘Fibroblast’



‘Inflammatory’



‘Miscellaneous’



“CRCHistoPhenotypes” dataset images

What are the right features?

What are the right features?

Depend on what you want to know!

What are the right features?

Depend on what you want to know!

- **Object: shape**
 - Local shape info, shading, shadows, texture

What are the right features?

Depend on what you want to know!

- **Object: shape**
 - Local shape info, shading, shadows, texture
- **Scene: geometric layout**
 - linear perspective, gradients, line segments

What are the right features?

Depend on what you want to know!

- **Object: shape**
 - Local shape info, shading, shadows, texture
- **Scene: geometric layout**
 - linear perspective, gradients, line segments
- **Material properties: albedo, feel, hardness**
 - Color, texture

What are the right features?

Depend on what you want to know!

- **Object: shape**
 - Local shape info, shading, shadows, texture
- **Scene: geometric layout**
 - linear perspective, gradients, line segments
- **Material properties: albedo, feel, hardness**
 - Color, texture
- **Action: motion**
 - Optical flow, tracked points

Image representations

- Templates
 - Intensity, gradients, etc.



Image
Intensity



Gradient
template

- Histograms
 - Color, texture, SIFT descriptors, etc.

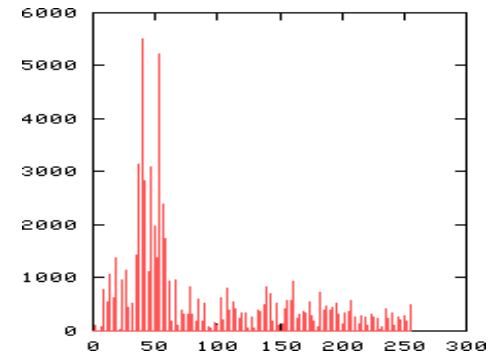
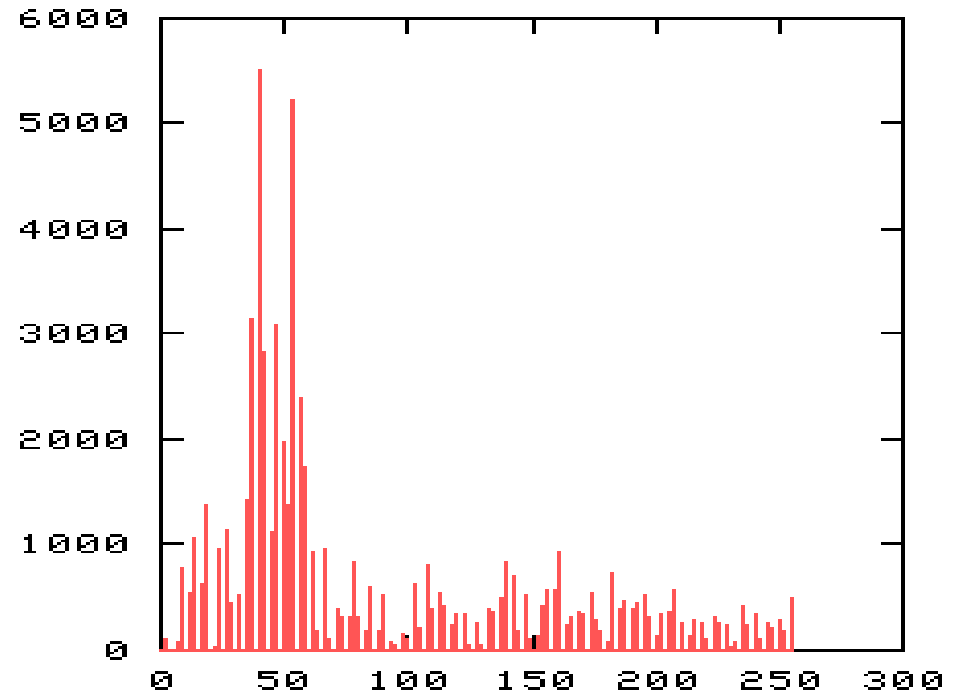
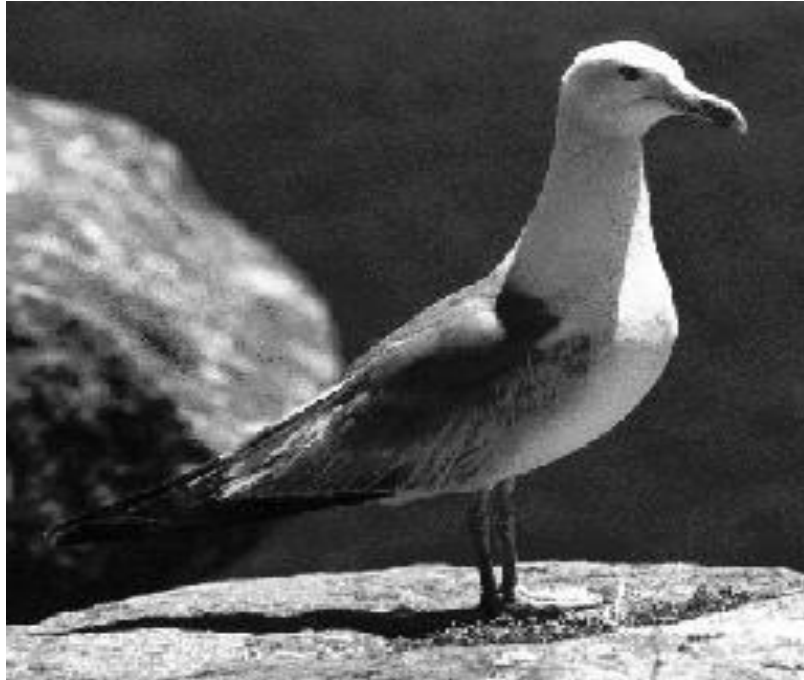


Image representations: histograms

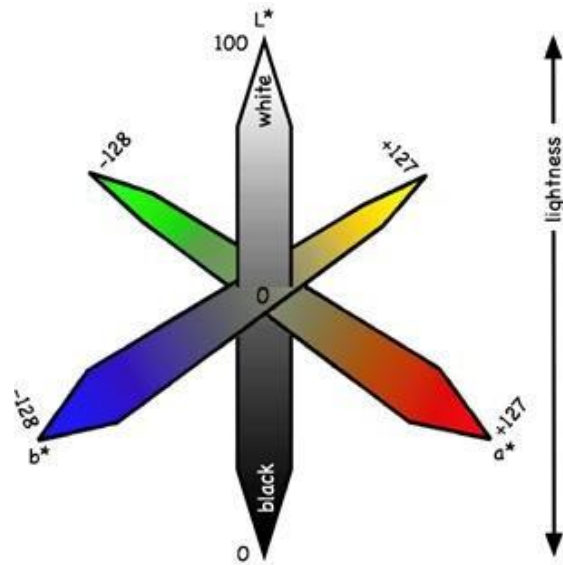


Global histogram

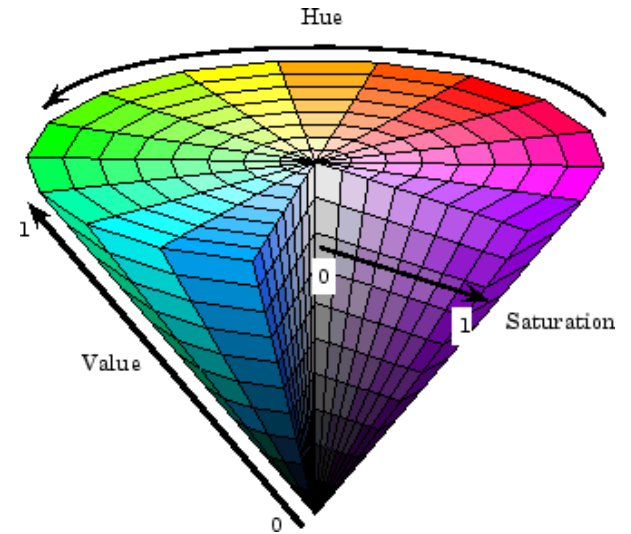
- Represent distribution of features
 - Color, texture, depth, ...

What kind of things do we compute histograms of?

- Color

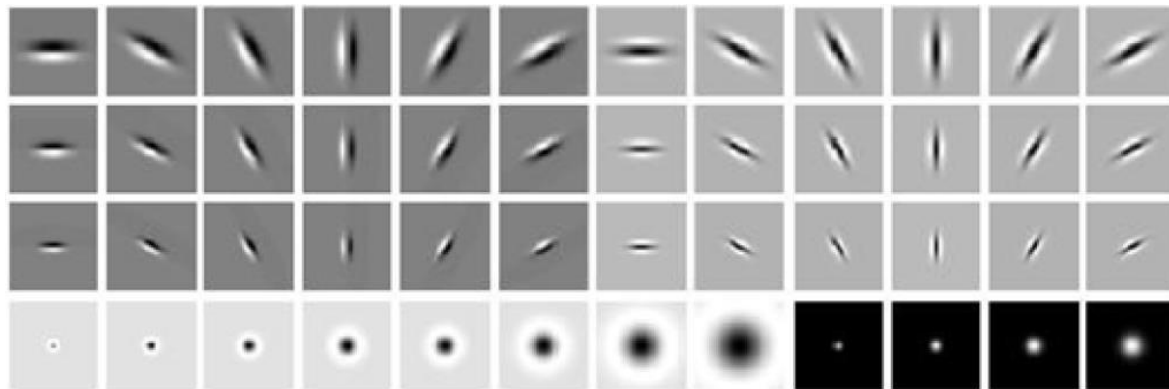


$L^*a^*b^*$ color space



HSV color space

- Texture (filter banks)



What kind of things do we compute histograms of?

Histograms of descriptors

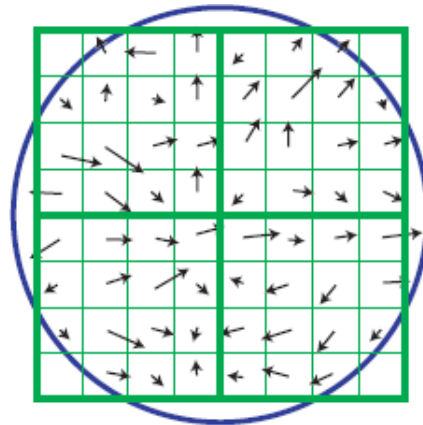
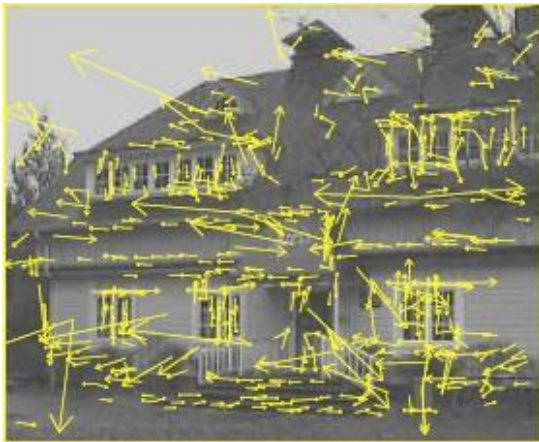
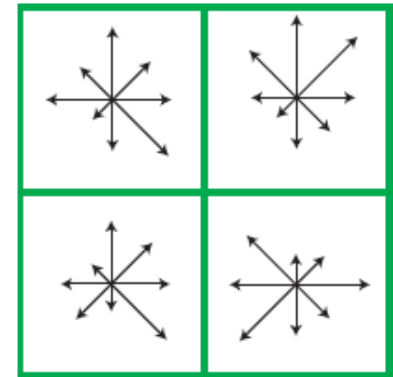


Image gradients

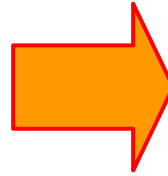


Keypoint descriptor

SIFT – [Lowe IJCV 2004]

What kind of things do we compute histograms of?

Bags of visual words



Bag of Words Model

SIFT features: Recap

- Can the same descriptors be used for matching different instances of the object?
 - Used in image classification, detection, etc.



Rigid transformations may not work here. Can we discard geometry for now, and bring it back in other ways?

Source: Yannis Avrithis, *Deep Learning for Vision*, Spring 2019 SIF

Analogy to documents

Of all the sensory impressions proceeding to the brain, the visual experiences are the dominant ones. Our perception of the world around us is based essentially on the messages that reach our eyes.

For a long time, the way in which the retinal image was processed by the visual centers in the brain was a mystery. It was only in the 1950s, when Hubel and Wiesel discovered the way in which the visual system works, that we began to know the truth about the process of perception.

Perception is a more complex process than we thought. It is following the path of the visual message to the various centers in the brain, Hubel and Wiesel have demonstrated that the message about the image falling on the retina undergoes a complex analysis in a system of nerve cells.

In this system each cell has its specific function and is responsible for a specific detail in the pattern of the retinal image.

**sensory, brain,
visual, perception,
retinal, cerebral cortex,
eye, cell, optical
nerve, image
Hubel, Wiesel**

China is forecasting a trade surplus of \$90bn (£51bn) to \$100bn this year, a threefold increase on 2004's \$32bn. The Commerce Ministry said the surplus would be created by a predicted 30% increase in exports to \$750bn, compared with \$580bn in 2004.

The increase in exports will annoy the US, which has long complained about China's trade policies. China's government has agreed to a deal with the US that the yuan is to be allowed to rise in value.

The government also needs to increase demand so that the yuan can rise in value. China has been allowed to trade within a narrow band, but the US wants the yuan to be allowed to rise freely. However, Beijing has made it clear that it will take its time and tread carefully before allowing the yuan to rise further in value.

**China, trade,
surplus, commerce,
exports, imports, US,
yuan, bank, domestic,
foreign, increase,
trade, value**

Bag-of-Words (BoW)

1) John likes to watch movies. Mary likes movies too.

2) Mary also likes to watch football games.

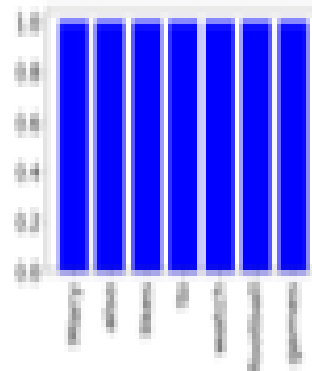
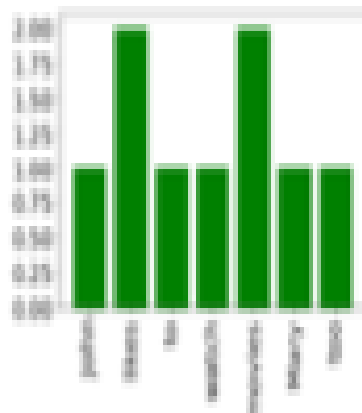
Samples



"John", "likes", "to", "watch", "movies", "Mary", "likes", "movies", "too"

"Mary", "also", "likes", "to", "watch", "football", "games"

Form Vocabulary



Histogram

Bag-of-Words (BoW)

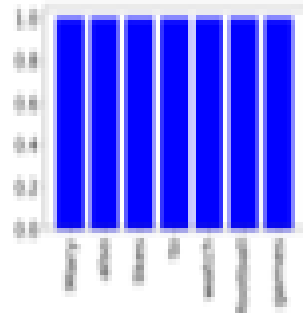
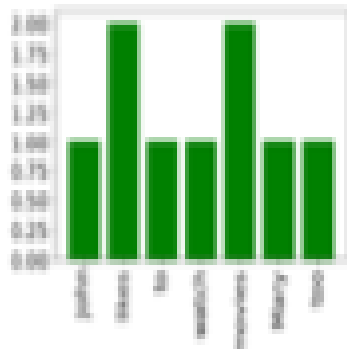
1) John likes to watch movies. Mary likes movies too.

2) Mary also likes to watch football games.



"John", "likes", "to", "watch", "movies", "Mary", "likes", "movies", "too"

"Mary", "also", "likes", "to", "watch", "football", "games"



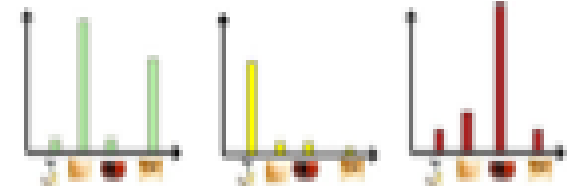
Samples



Form Vocabulary



Histogram



- Use Bow for Retrieval, Classification, etc.

Bag-of-Words (BoW) Retrieval



query

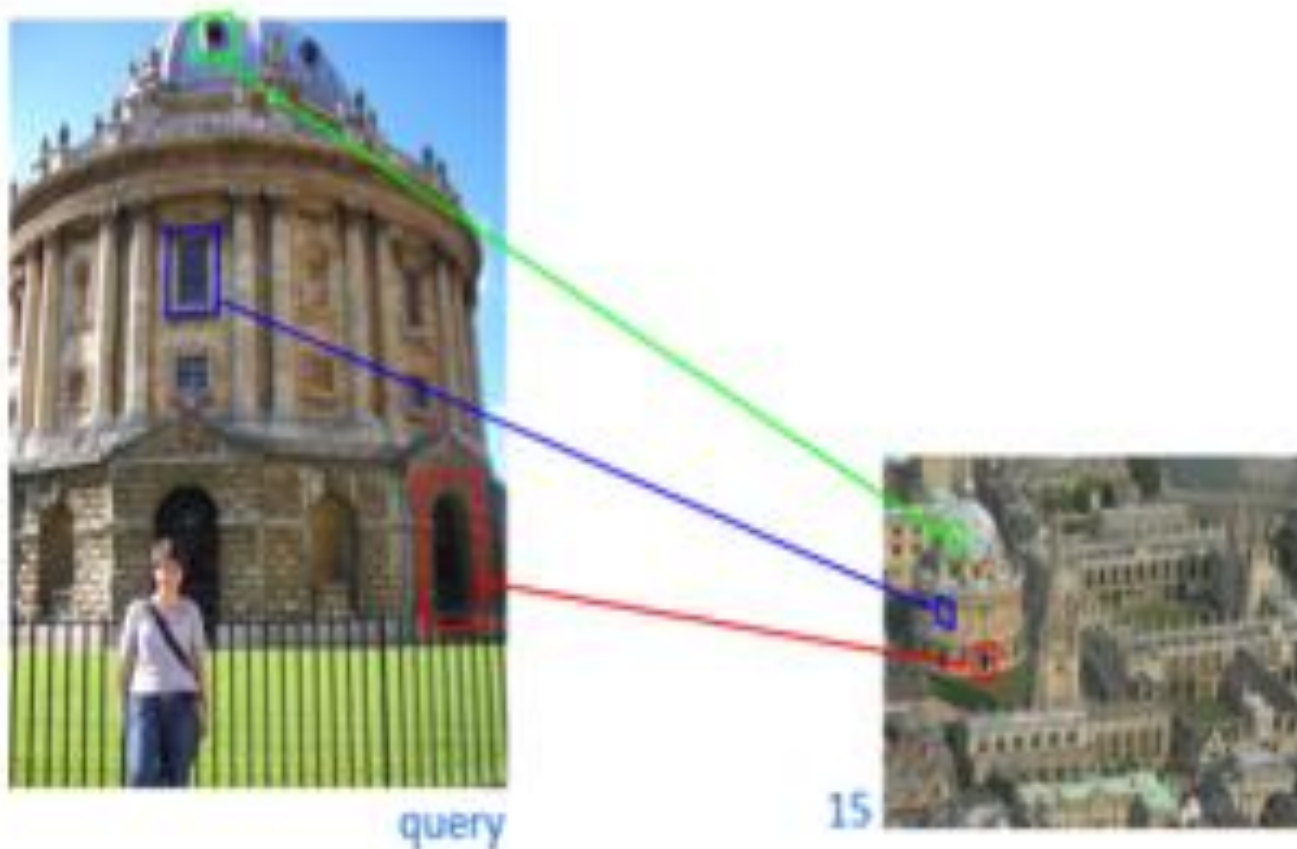


15

Query vs dataset image

¹Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

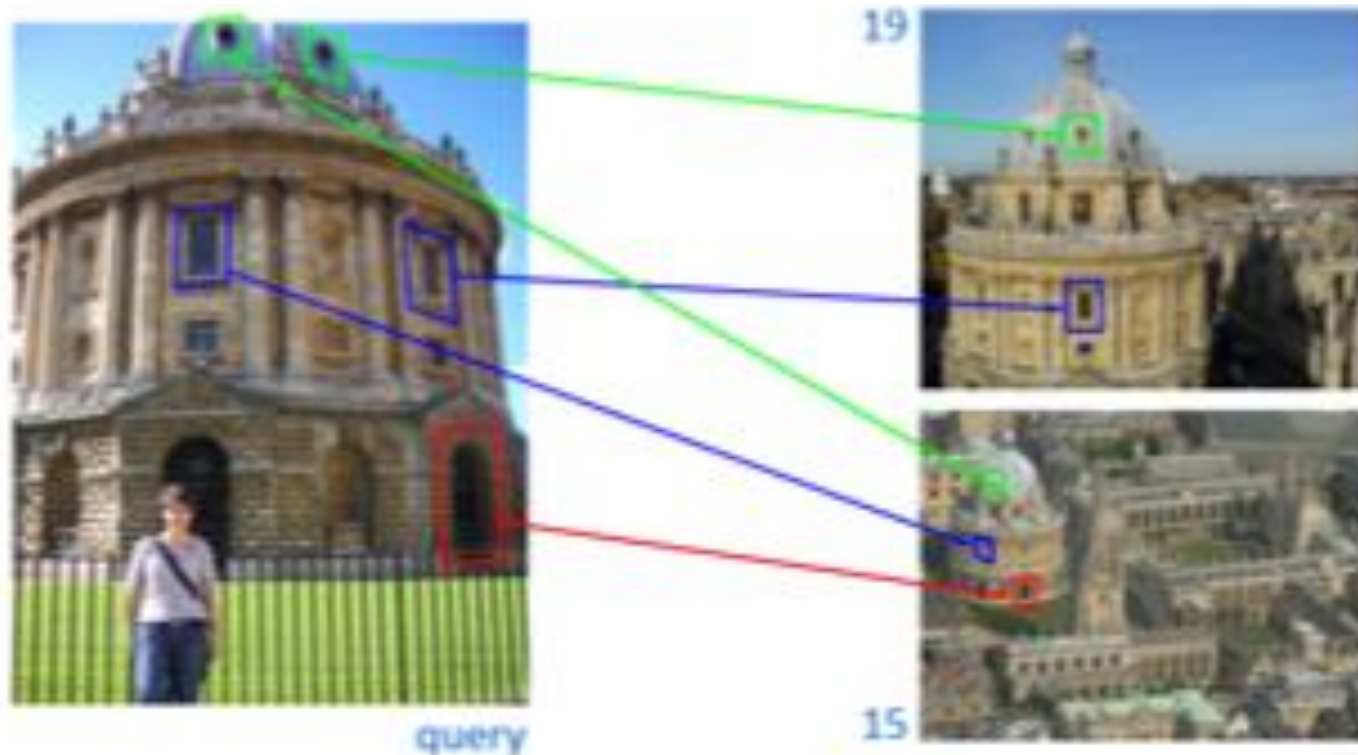
Pairwise Key Points Matching



Pairwise descriptor matching

¹Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

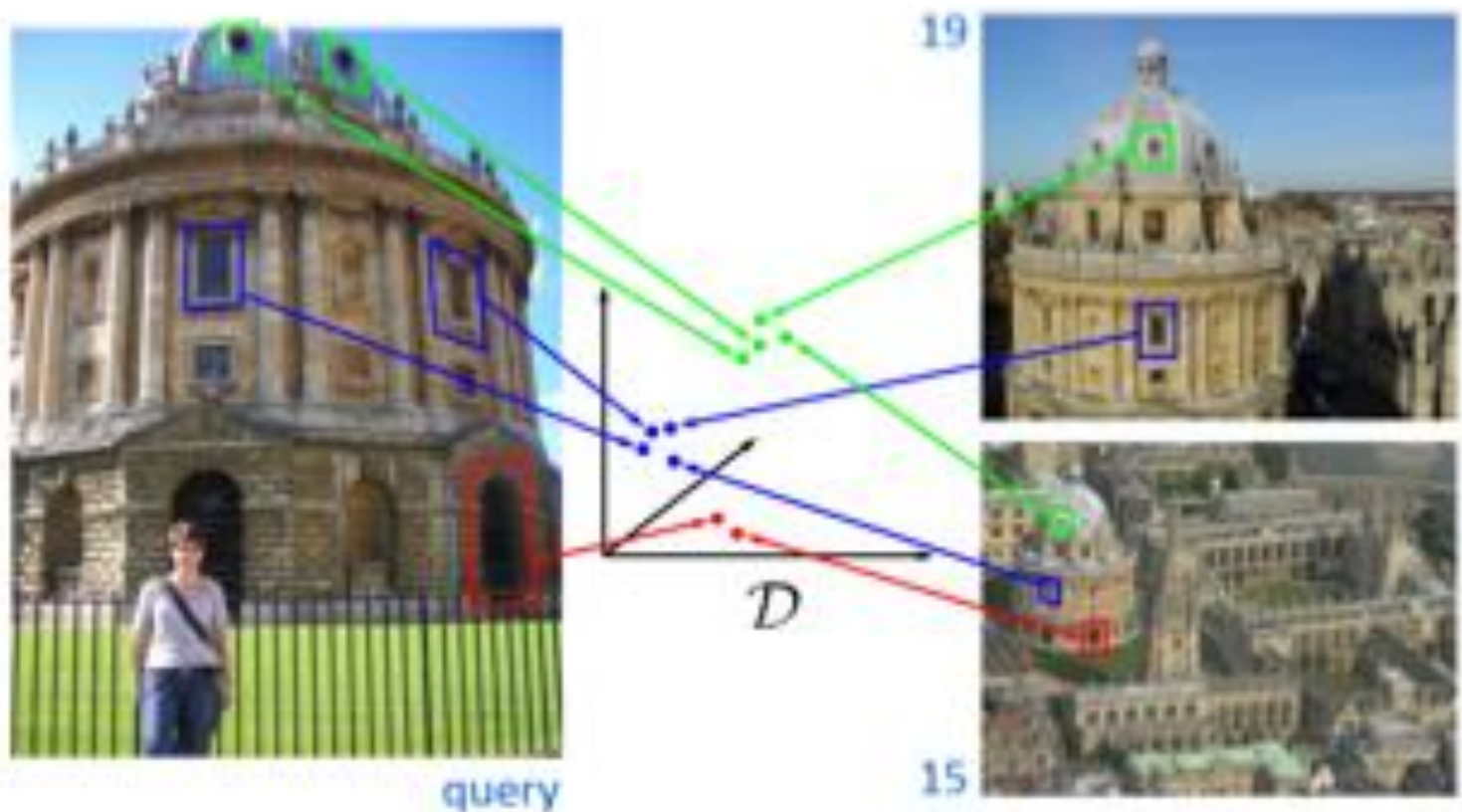
Pairwise Key Points Matching



Pairwise descriptor matching for **every** dataset image

¹Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

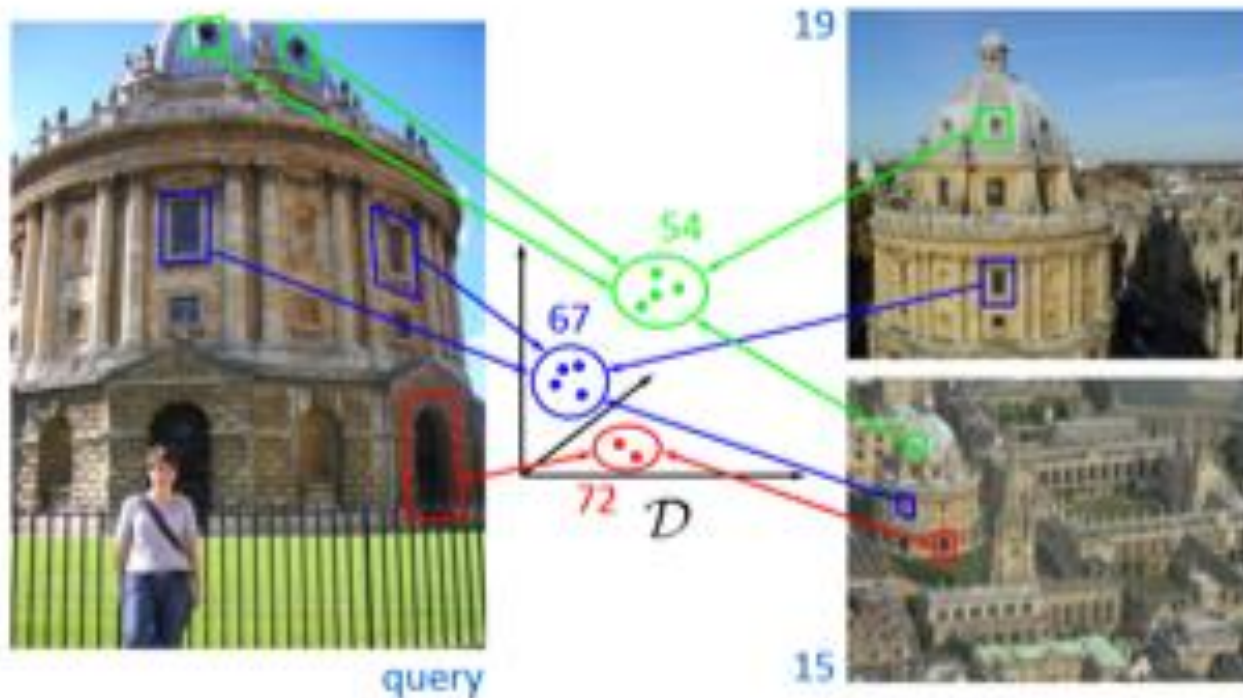
How to Resolve this ?



Similar descriptors should all be nearby in the descriptor space

¹Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

Bag-of-Words (BoW) Retrieval

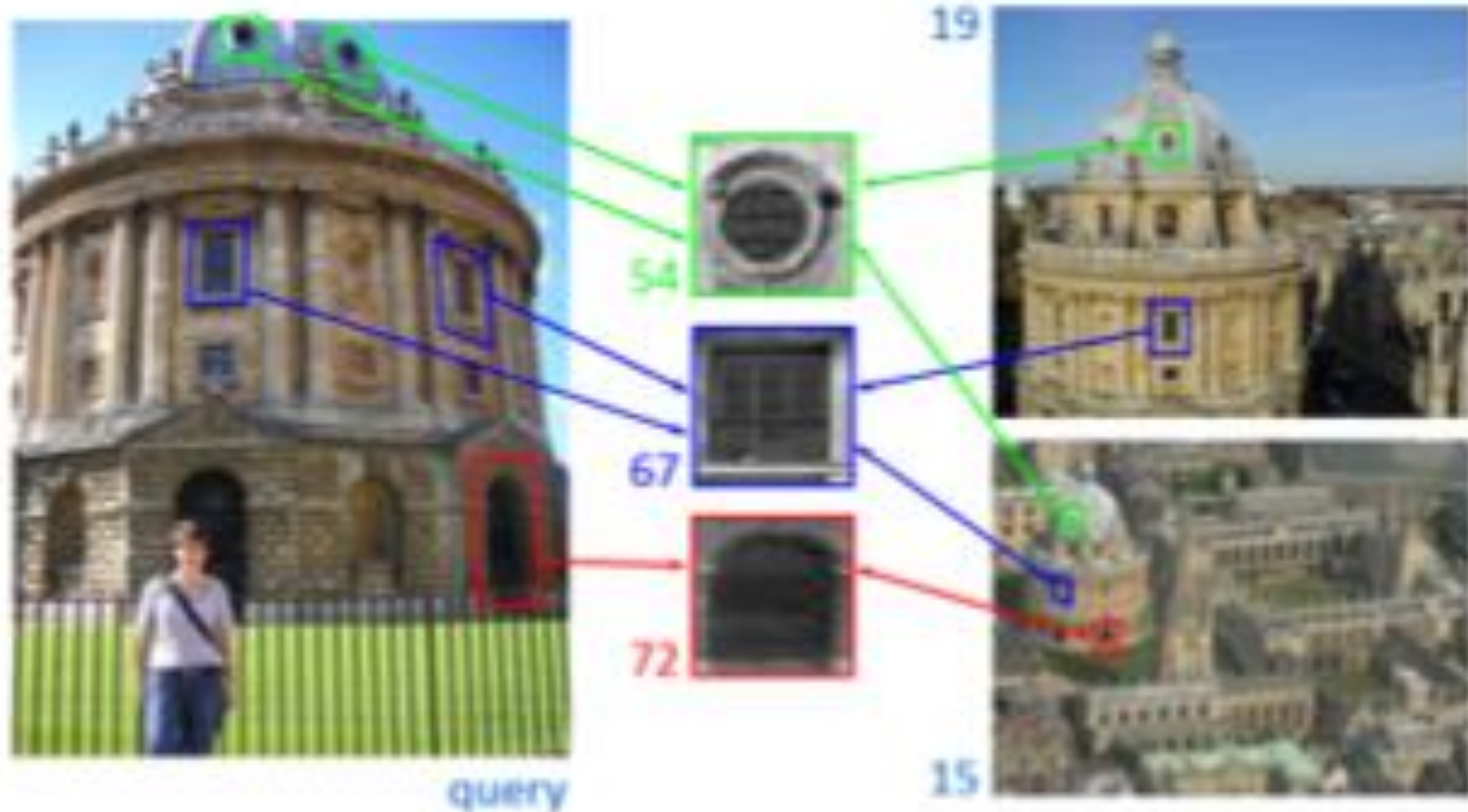


Quantize them into visual words

¹Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

Bag-of-Words (BoW) Retrieval

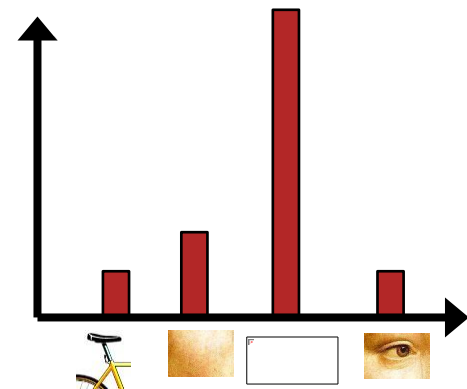
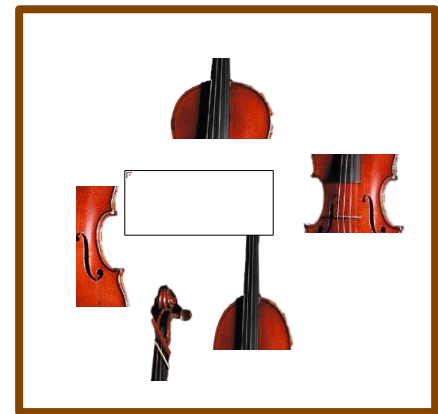
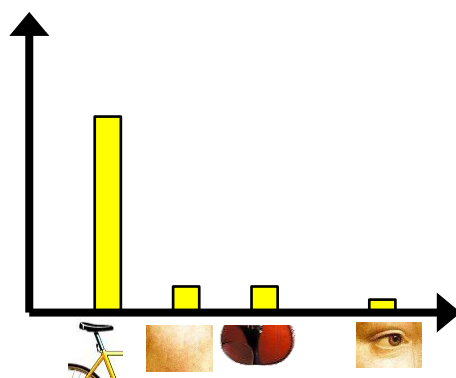
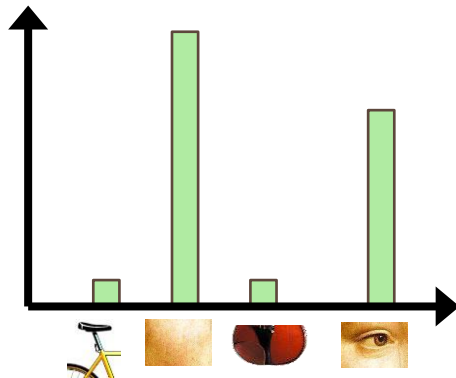
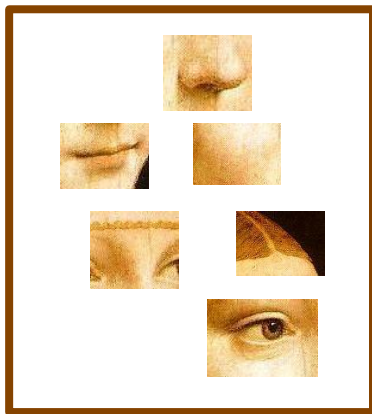
- Come up with a common representation for the common descriptors.
- Take mean of common visual words in the entire repository (offline)



Now visual words act as a proxy. No pairwise matching needed

Bags-of-visual-words

1. Extract local features
2. Learn “visual vocabulary”
3. Quantize local features using visual vocabulary
4. Represent images by frequencies of “visual words”



BoW Retrieval

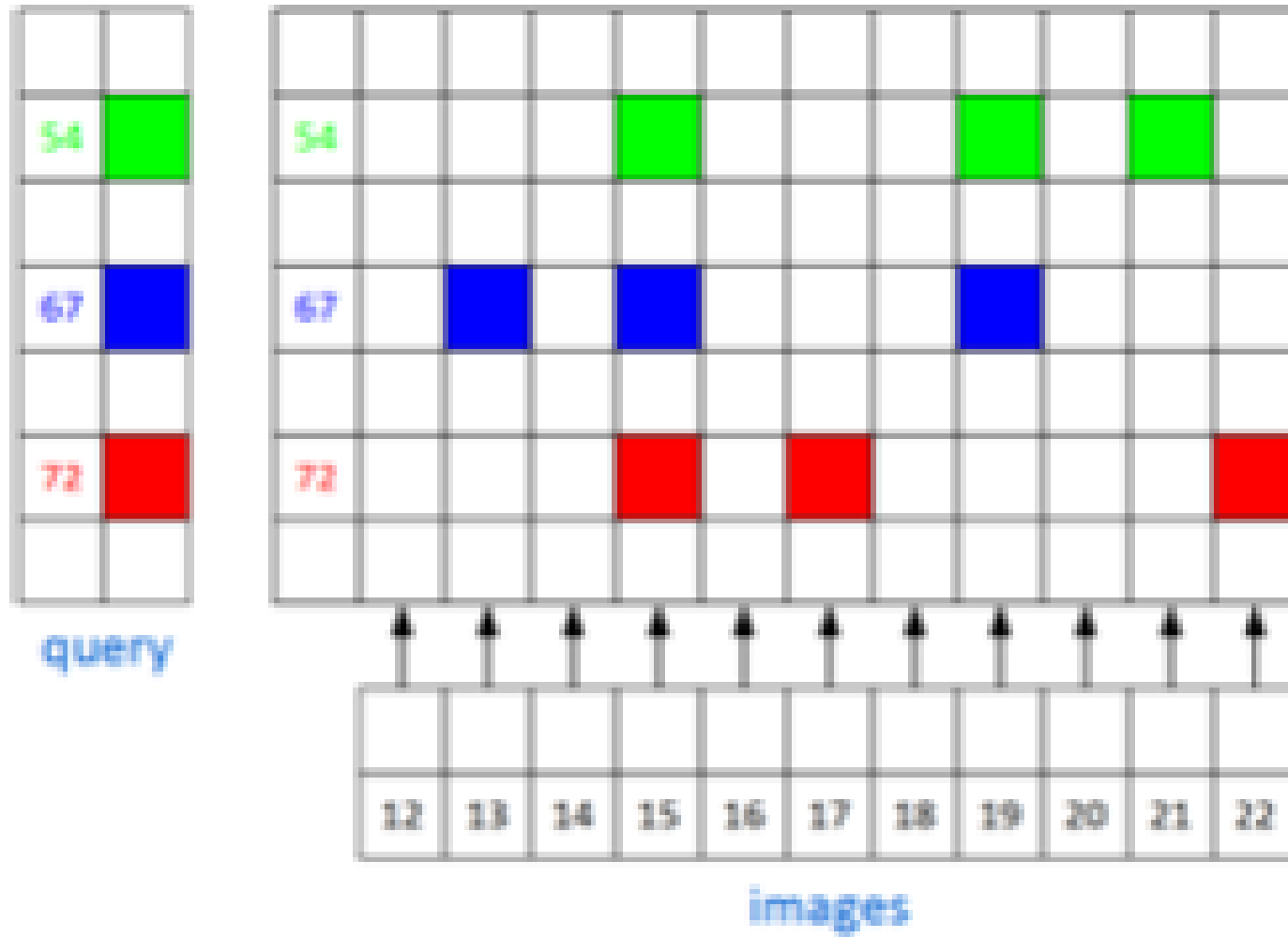
- Each image is represented by a vector $\mathbf{z} \in \mathbb{R}^k$, where k is size of codebook
 - Each element $z_i = w_i n_i$ where w_i is a fixed weight per visual word and n_i number of occurrences of this word in the image
- Given a set of n images represented by matrix $Z \in \mathbb{R}^{k \times n}$ (each image as a column) and query image $\mathbf{q}$, we need a vector of similarities:

$$\mathbf{s} = S_{\text{BoW}}(Z, \mathbf{q}) := Z^T \mathbf{q}$$

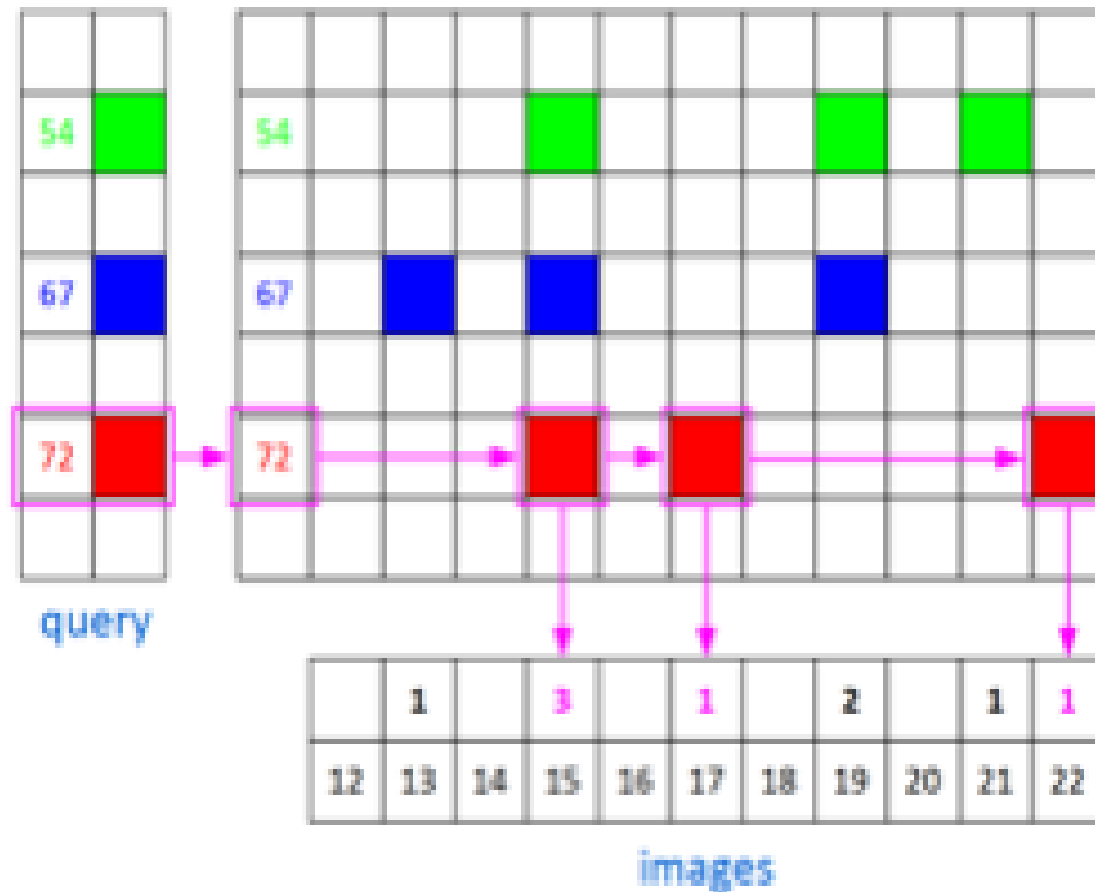
and then sort $\mathbf{s}$ by descending order

- Note: With L_2 -normalization, this is equivalent to measuring Euclidean distance: for vectors $\mathbf{z}$ and $\mathbf{q}$, $\|\mathbf{z} - \mathbf{q}\|^2 = 2(1 - \mathbf{z}^T \mathbf{q})$
- When $k \gg p$, where p is the number of features per image on average, Z and $\mathbf{q}$ are sparse
- Rather than check whether a word is contained in an image, check **which images contain a given word**

BoW Retrieval



BoW Retrieval



²Sivic and Zisserman, Video Google: A Text Retrieval Approach to Object Matching in videos, ICCV 2003

BoW Retrieval

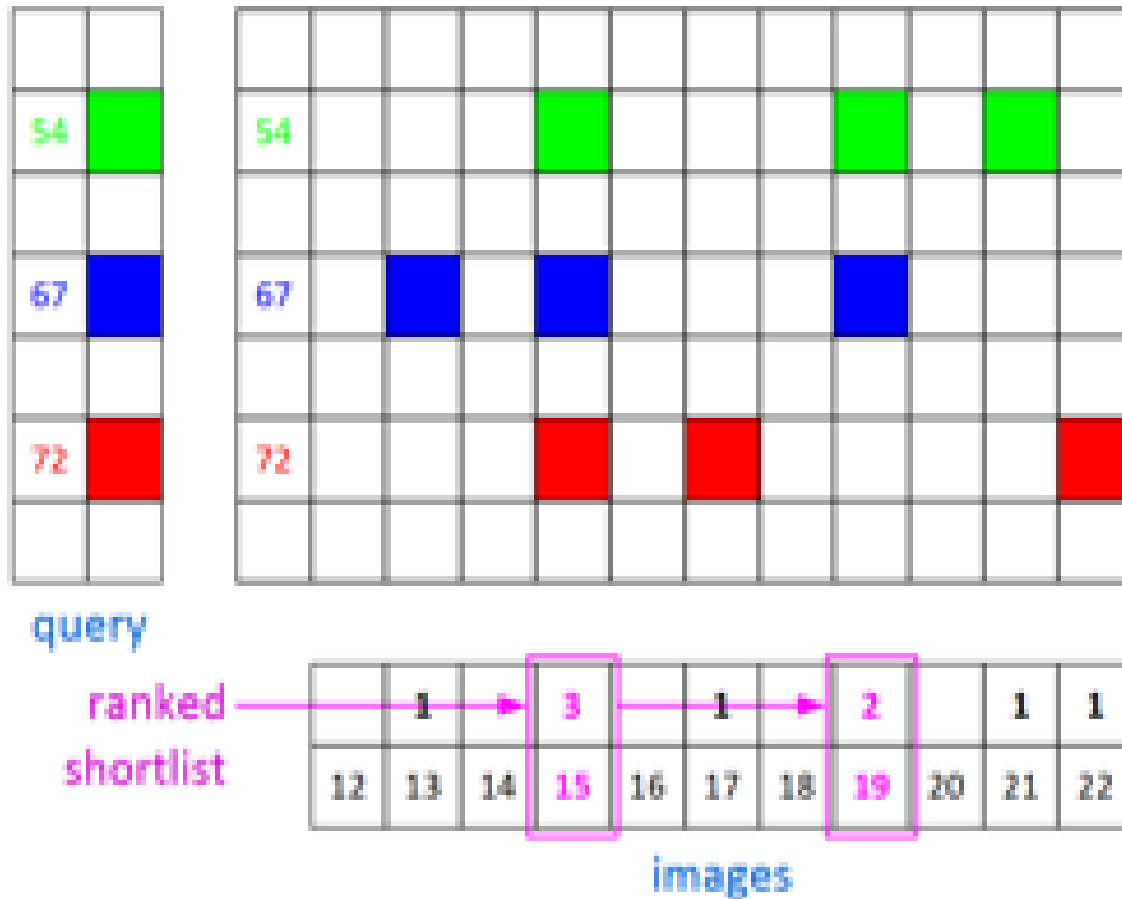


Image categorization with bag of words

Training

1. Extract visual words and descriptors for all training images
2. Cluster descriptors
3. Quantize descriptors using cluster centers to get “visual words”
4. Represent each image by normalized counts of “visual words”
5. Train classifier on labeled examples using histogram values as features

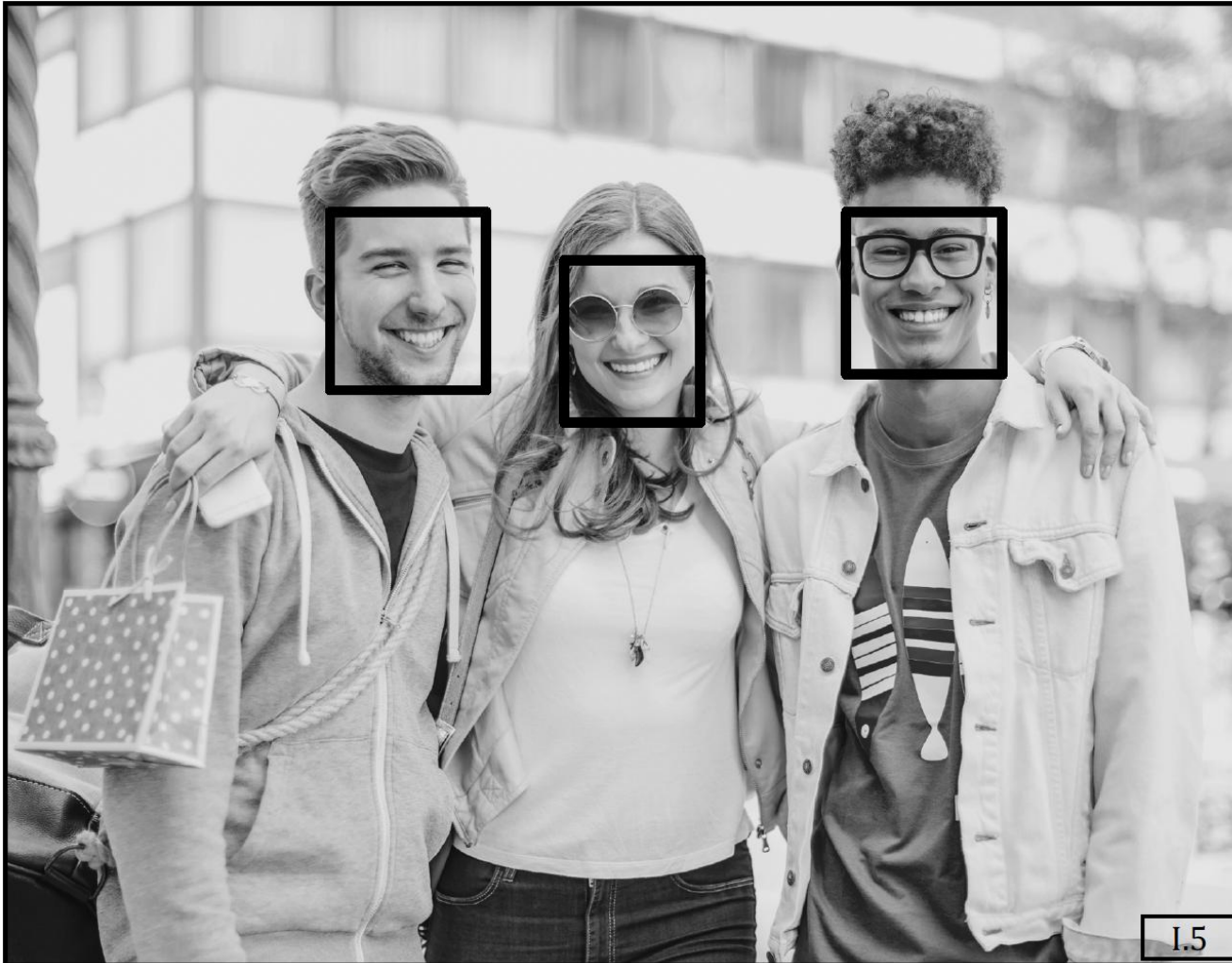
Testing

1. Extract keypoints/descriptors and quantize into visual words
2. Compute visual word histogram
3. Compute label or confidence using classifier

Face Detection

Face Detection

- Locate Human Faces in Images



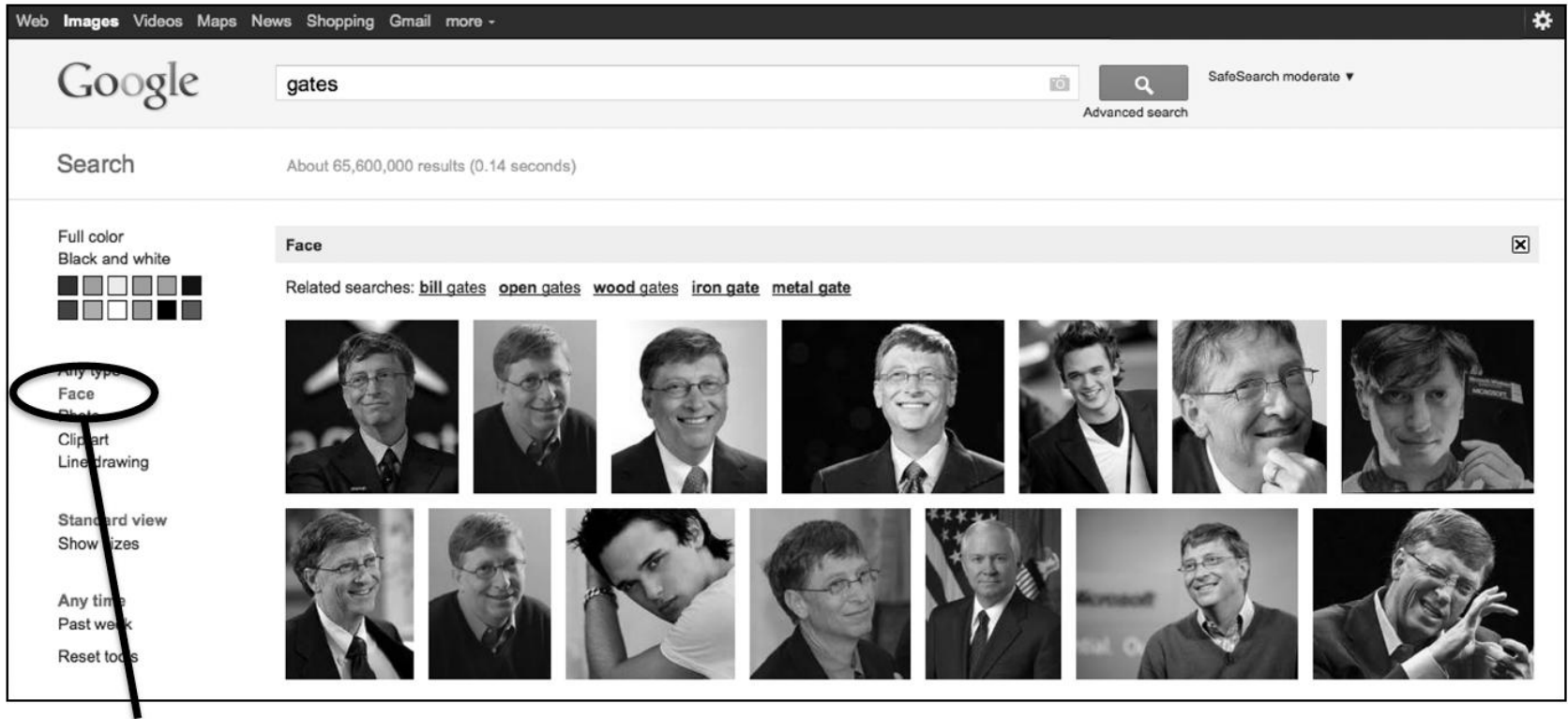
Where Face Detection is Used ?



Automatic Selection of Camera Settings
(Autofocus, Exposure, Color Balance, etc.)

Where Face Detection is Used ?

- Google image search

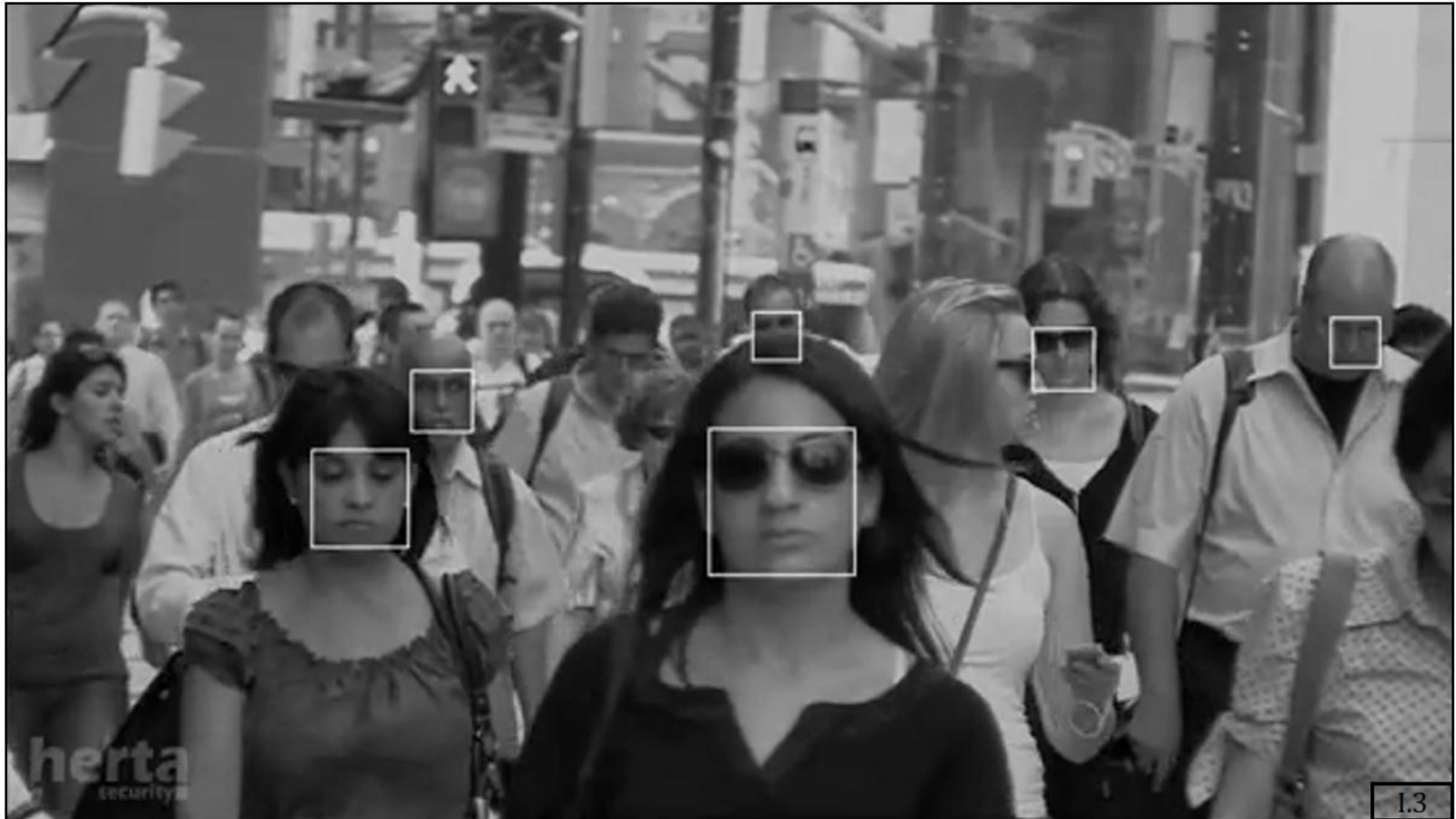


Face Detection

Faces of people named "Gates"

Finding People using Search Engines

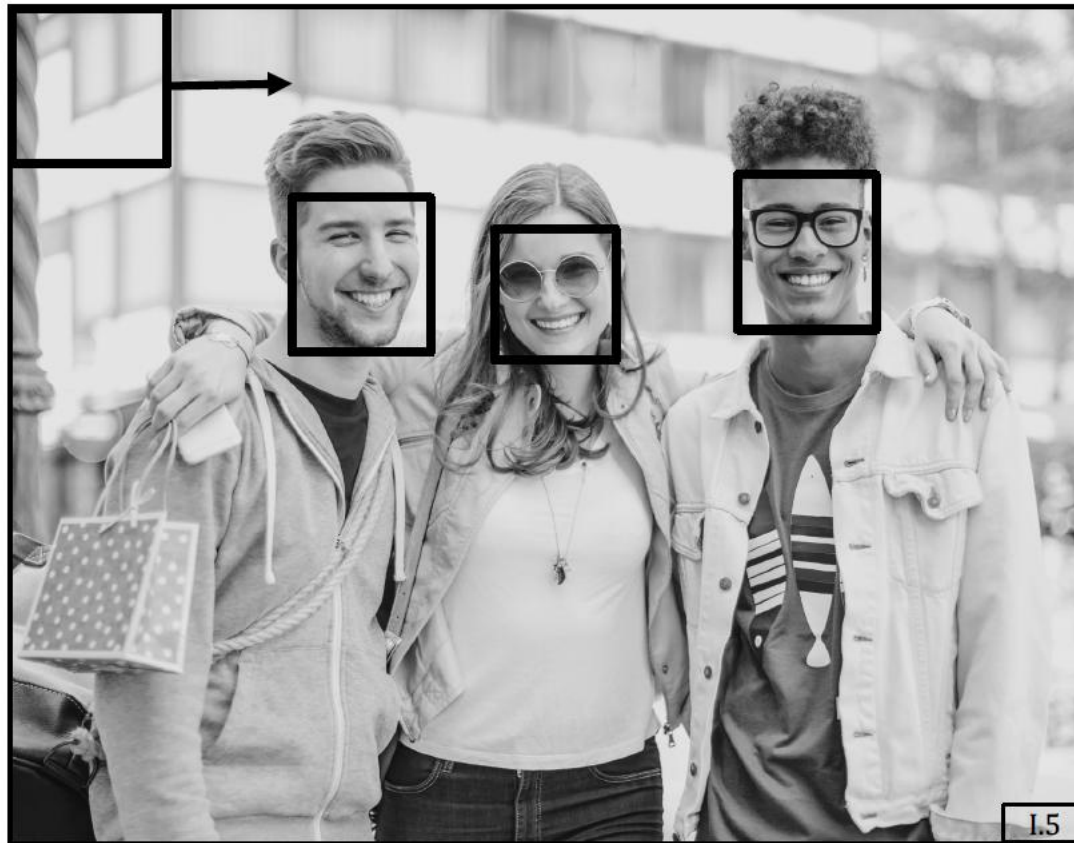
Where Face Detection is Used ?



Biometrics, Surveillance, Monitoring

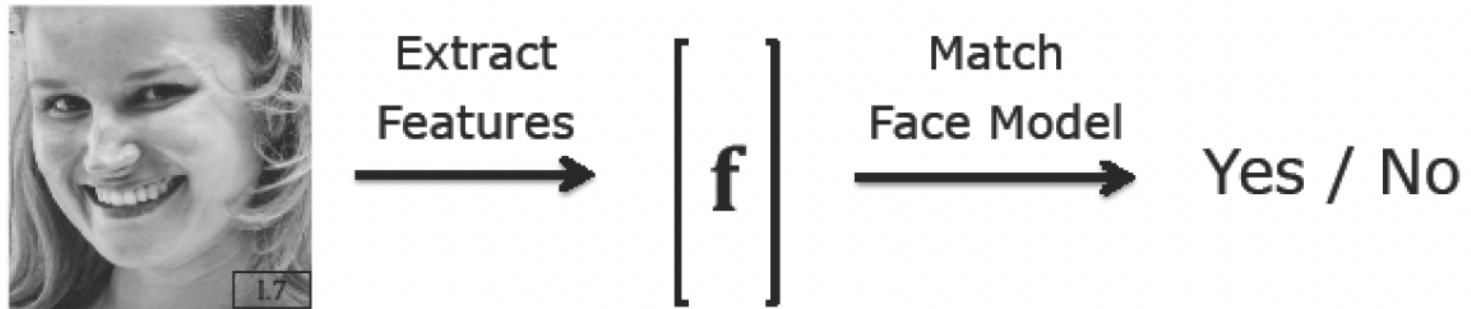
Face Detection is Computers

Slide windows of different sizes across image.
At each location match window to face model.



Face Detection Framework

For each window:



Features: Which features represent faces well?

Classifier: How to construct a face model and efficiently classify features as face or not?

What are Good Features?

Interest Points (Edges, Corners, SIFT)?



Facial Components (Templates)?



Characteristics of Good Features

Discriminate Face/Non-Face



Extremely Fast to Compute

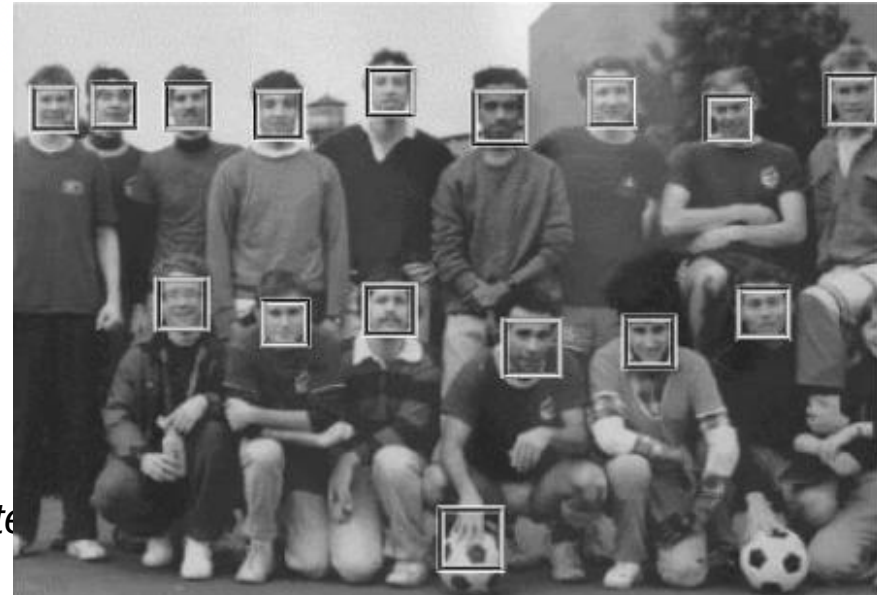
Need to evaluate millions of windows in an image

Viola-Jones Face Detection Algorithm

- Detection in the Pre-Deep Learning Era

Viola-Jones Algorithm

- A framework to perform object detection in real-time
- Used predominantly for detecting frontal upright faces.
- Consists of three main components:
 - Weak classification using **Haar-like features** and **Integral Images**
 - Adaboost to build strong classifier
 - Classifier cascades



• Viola, Jones, *Rapid Object Detection using a Boosted Cascade of Simple Features*, CVPR 2001

Haar Features

Set of Correlation Responses to Haar Filters

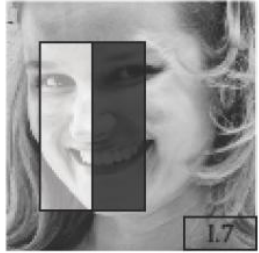


Input Image

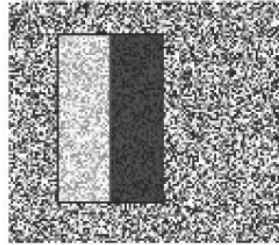
$$\begin{array}{c}
 \begin{bmatrix}
 \begin{array}{|c|c|} \hline \text{white} & \text{black} \\ \hline \end{array} \\
 \begin{array}{|c|c|} \hline \text{black} & \text{white} \\ \hline \end{array} \\
 \begin{array}{|c|c|} \hline \text{white} & \text{black} \\ \hline \end{array} \\
 \begin{array}{|c|c|} \hline \text{white} & \text{black} \\ \hline \end{array} \\
 \vdots
 \end{bmatrix}
 \begin{array}{l}
 H_A \\
 H_B \\
 H_C \\
 H_D \\
 \\
 \end{array}
 \otimes
 =
 \begin{bmatrix}
 V_A[i,j] \\
 V_B[i,j] \\
 V_C[i,j] \\
 V_D[i,j] \\
 \vdots
 \end{bmatrix}
 \end{array}$$

Haar Filters
Haar Features
 $\mathbf{f}[i,j]$

Discriminative Ability of Haar Features



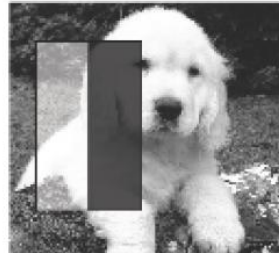
$$V_A = 64$$



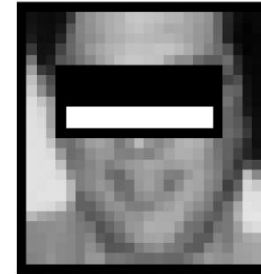
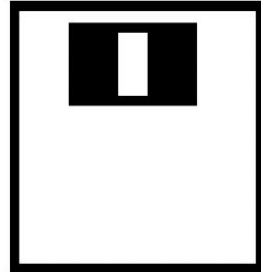
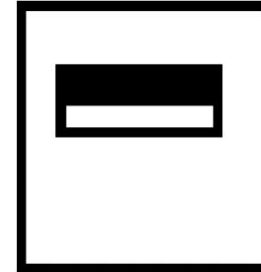
$$V_A \approx 0$$



$$V_A = 16$$



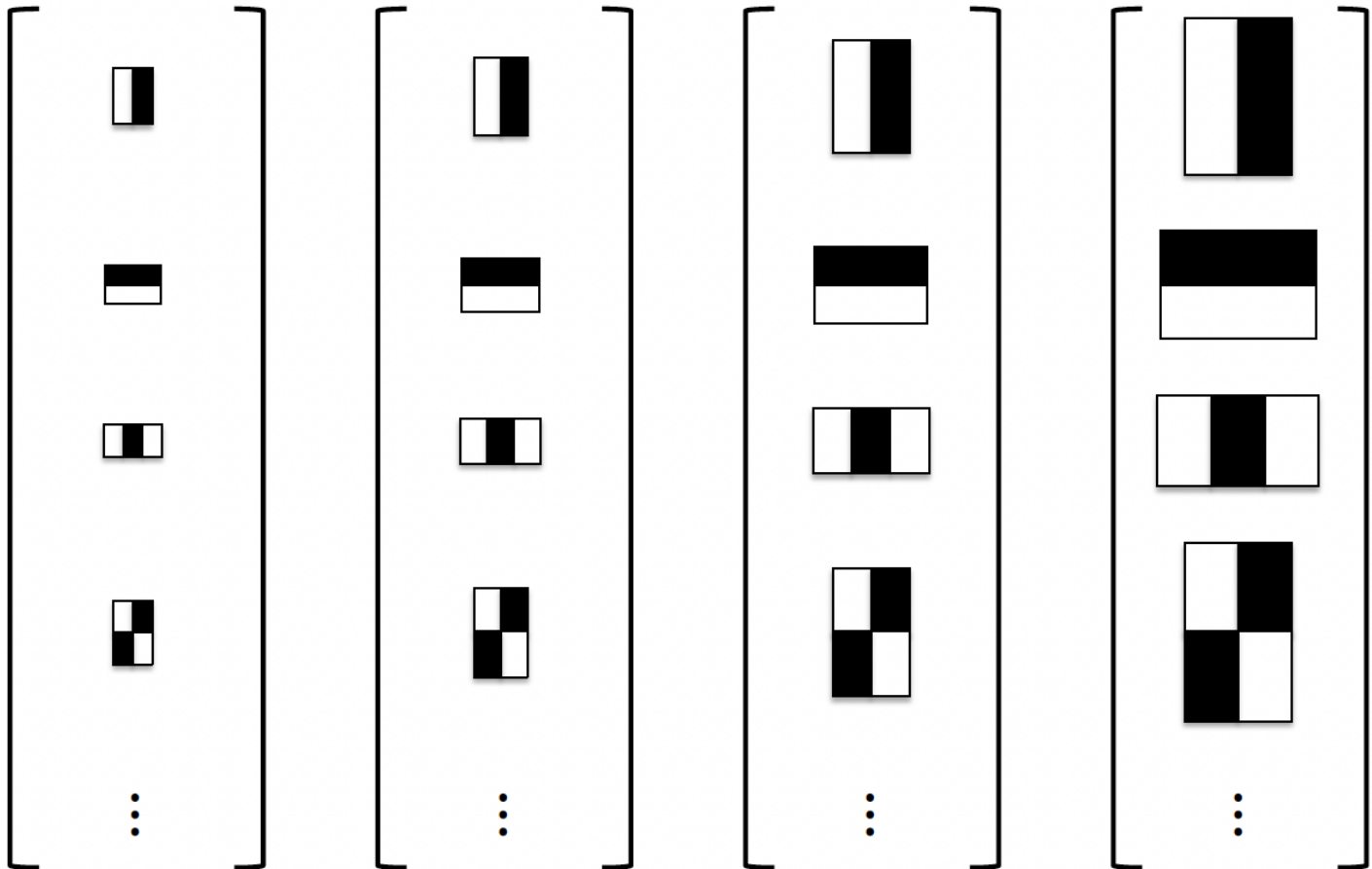
$$V_A = -127$$



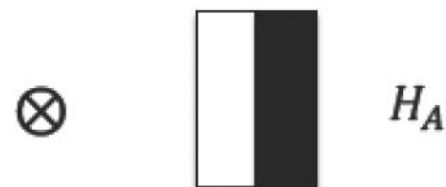
- Haar features are sensitive to directionality patterns

Detecting Faces of Different Size

Compute Haar Features at different scales to detect faces of different sizes.



Computing Haar Features



White = 1, Black = -1

Response to Filter H_A at location (i, j) :

$$V_A[i, j] = \sum_m \sum_n I[m, n] H_A[m - i, n - j] \quad \boxed{1}$$

$$V_A[i, j] = \sum (\text{pixel intensities in white area}) \\ - \sum (\text{pixels intensities in black area})$$

Haar Features: Computational Costs



$$Value = \sum(\text{pixel intensities in white}) - \sum(\text{pixel intensities in black})$$

Computation cost = $(N \times M - 1)$ additions per pixel per filter per scale

Can We Do Better?

Integral Images

A table that holds the sum of all pixel values to the left and top of a given pixel, inclusive.

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
97	113	147	108	125	142
95	111	168	122	130	137
96	104	172	130	126	130

Image *I*

98	208	329	454	576	705
197	417	658	899	1137	1395
294	623	988	1340	1701	2093
392	833	1330	1790	2274	2799
489	1043	1687	2255	2864	3531
584	1249	2061	2751	3490	4294
680	1449	2433	3253	4118	5052

Integral Image *II*

Summation within a Rectangle

Fast summations of arbitrary rectangles using integral images

2

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
97	113	147	103	125	142
95	111	168	122	120	137
96	104	172	130	126	130

Image *I*

98	208	329	454	576	705
107	417	658	899	1137	1395
294	623	988	1340	1701	2093
392	833	1330	1790	2274	2799
489	1043	1687	2255	2864	3531
584	1249	2061	2751	3490	4204
680	1449	2433	3253	4118	5052

Integral Image *II*

$$\begin{aligned}
 Sum &= II_P - II_Q - II_S + II_R \\
 &= 3490 - 1137 - 1249 + 417 = 1521
 \end{aligned}$$

Computational Cost: Only 3 additions

Haar Response using Integral Image

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
97	113	147	108	125	142
95	111	168	122	130	137
96	104	172	130	126	130

Image I

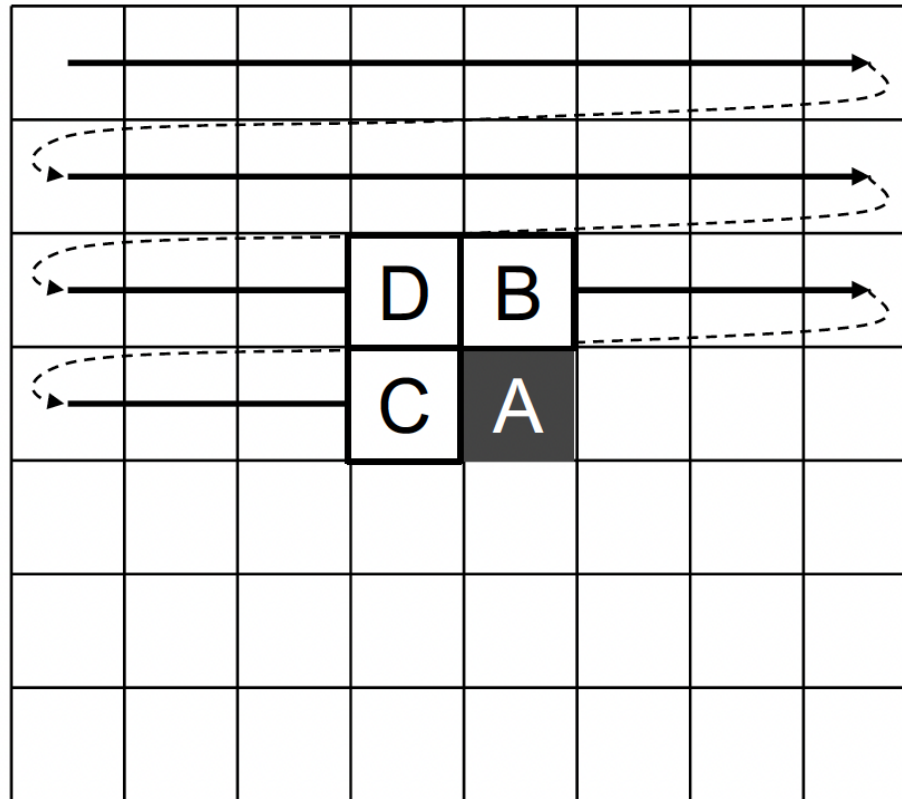
98	208	329	454	576	705
197	417	658	899	1137	1395
294	623	988	1340	1701	2093
392	833	1330	1790	2274	2799
489	1043	1687	2255	2864	3531
584	1249	2061	2751	3490	4294
680	1449	2433	3253	4118	5052

Integral Image II

$$\begin{aligned}
 V_A &= \sum(\text{pixel intensities in white}) - \sum(\text{pixel intensities in black}) \\
 &= (II_O - II_T + II_R - II_S) - (II_P - II_Q + II_T - II_O) \quad \boxed{1} \\
 &= (2061 - 329 + 98 - 584) - (3490 - 576 + 329 - 2061) = 64
 \end{aligned}$$

Computational Cost: Only 7 additions

Computing Integral Image



Raster
Scanning

Let I_A and II_A be the values of Image and Integral Image, respectively, at pixel A .

$$II_A = II_B + II_C - II_D + I_A$$

Haar Features using Integral Images

Integral image needs to be computed once per test image.
Allows fast computations of Haar features.



Input Image

$$\begin{matrix} \otimes & \begin{bmatrix} \begin{array}{|c|c|} \hline \text{white} & \text{black} \\ \hline \end{array} & H_A \\ \begin{array}{|c|c|} \hline \text{black} & \text{white} \\ \hline \end{array} & H_B \\ \begin{array}{|c|c|} \hline \text{white} & \text{black} & \text{white} \\ \hline \end{array} & H_C \\ \begin{array}{|c|c|} \hline \text{white} & \text{black} \\ \hline \text{black} & \text{white} \\ \hline \end{array} & H_D \\ \vdots & \end{bmatrix} & = & \begin{bmatrix} V_A[i,j] \\ V_B[i,j] \\ V_C[i,j] \\ V_D[i,j] \\ \vdots \end{bmatrix} \\ \text{Haar Filters} & & \text{Haar Features} \end{matrix}$$

Weak Classifier

- Each feature is considered as a weak classifier
- Given feature f_j , a threshold θ_j and a parity p_j indicating the direction of the inequality sign, classifier can be defined as:

$$h_j(x) = \begin{cases} 1 & \text{if } p_j f_j(x) < p_j \theta_j \\ 0 & \text{otherwise} \end{cases}$$

- Each classifier weak on its own, but combination gives strong classifiers

Do you see any problem here?

For a sliding window base detector of size 24×24 , there exist over 160k features. How to identify the best ones?

Classifier Training using AdaBoost

Recall: AdaBoost learns a strong classifier as a linear combination of weak classifiers

- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{m}, \frac{1}{l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \dots, T$:

1. Normalize the weights,

$$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$$

so that w_t is a probability distribution.

2. For each feature, j , train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t , $\epsilon_j = \sum_i w_i |h_j(x_i) - y_i|$.

3. Choose the classifier, h_t , with the lowest error ϵ_t .
4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_i^{1-\epsilon_i}$$

where $\epsilon_i = 0$ if example x_i is classified correctly, $\epsilon_i = 1$ otherwise, and $\beta_i = \frac{e^{-\epsilon_i}}{\sum_{j=1}^n e^{-\epsilon_j}}$.

- The final strong classifier is:

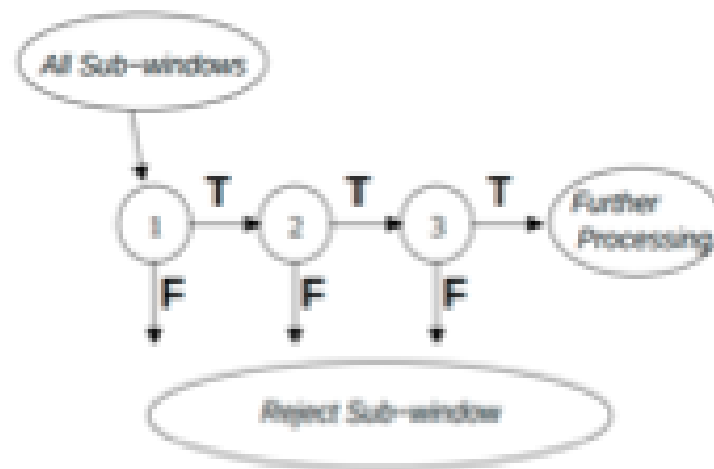
$$h(x) = \begin{cases} 1 & \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

Classifier Cascade

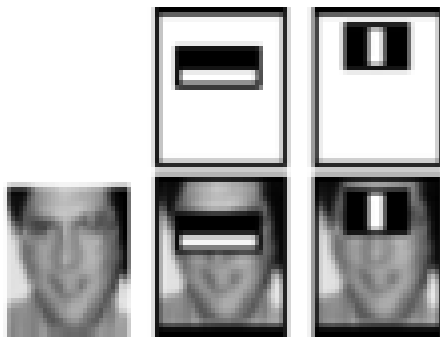
Intuition:

- Very less number of sub-windows actually have objects (False positive rate very high)
- But computation is equal across all sub-windows
- Eliminate negative windows earlier thereby, reducing computation time spent on them

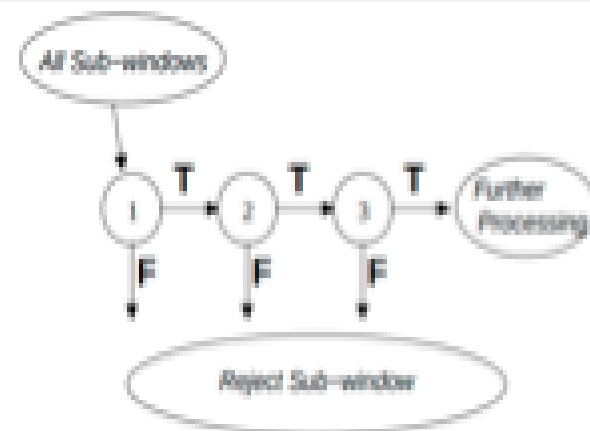


Classifier Cascade: Why does it matter ?

- A 1 MP has $\sim 10^6$ pixels and a comparable number of candidate face locations $\Rightarrow$ our ideal false positive (FP) rate has to be less than 10^{-6}



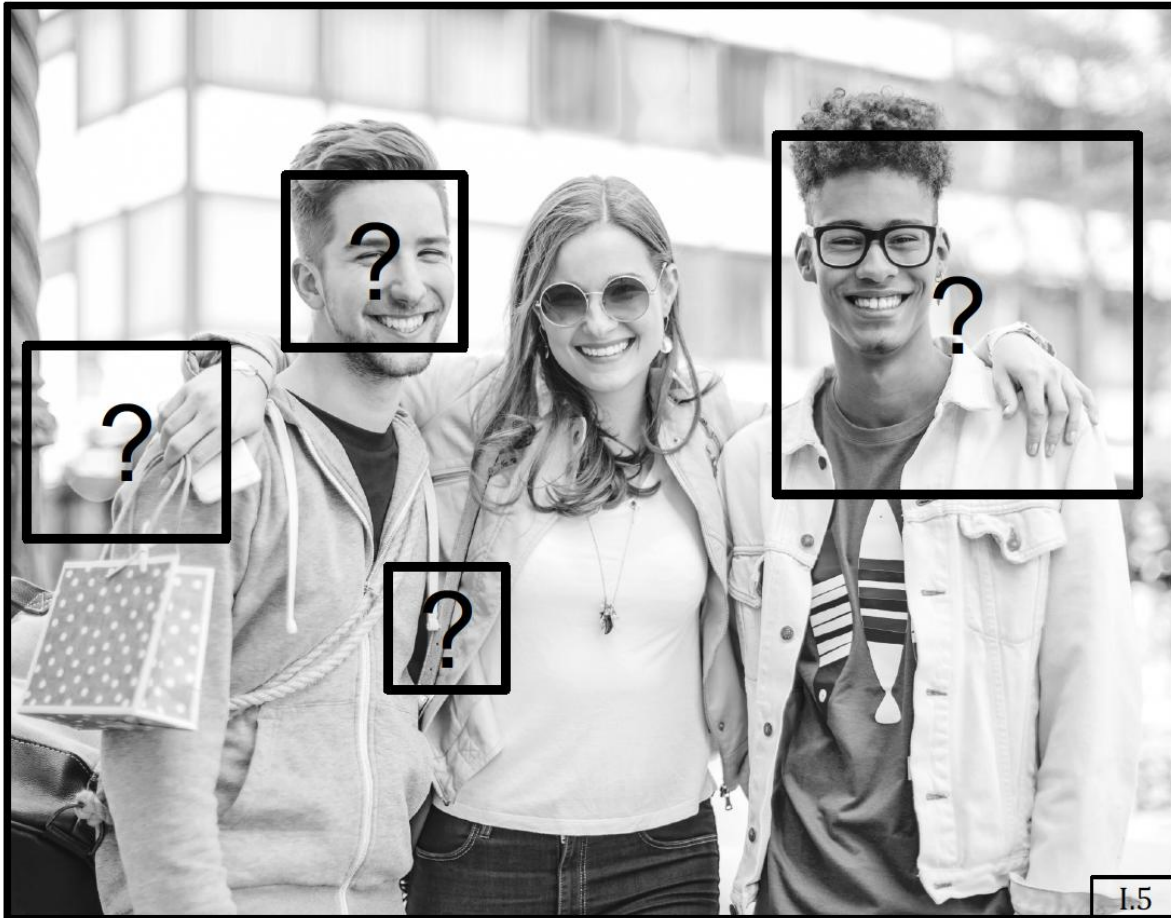
- This feature combination can yield 100% detection rate and 50% FP rate
- A 200-feature classifier can yield 95% detection rate and FP rate of 1 in 14084. Not enough!



- Detection rate and FP rate of cascade are found by multiplying respective rates of individual stages
- Detection rate of 0.9 and FP rate on the order of 10^{-6} can be achieved by a 10-stage cascade if each stage has detection rate of 0.99 ($0.99^{10} \approx 0.9$) and FP rate of about 0.30 ($0.3^{10} \approx 6 \times 10^{-6}$)

Classifier for Face Detection

Given the features for a window, how to decide whether it contains a face or not?

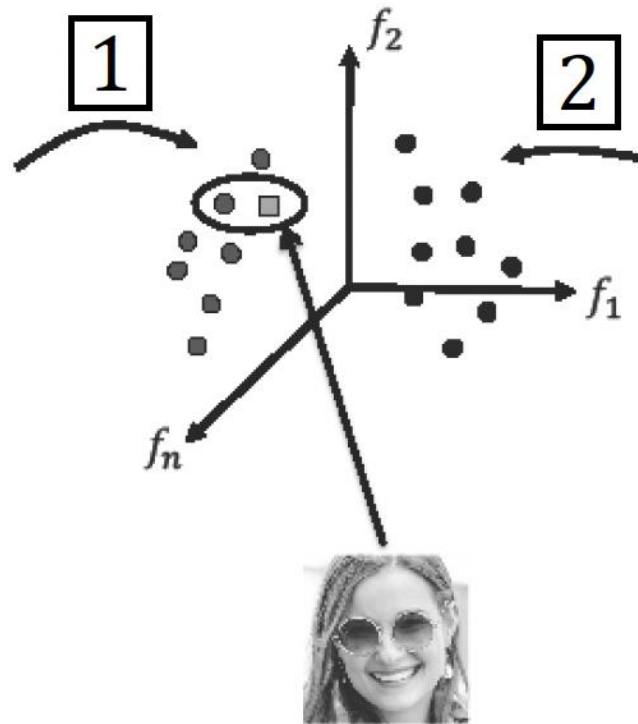


Nearest Neighbour Classifier

Find the nearest training sample using $\mathcal{L}^2$ distance and assign its label.



Training Data
of Face



Face



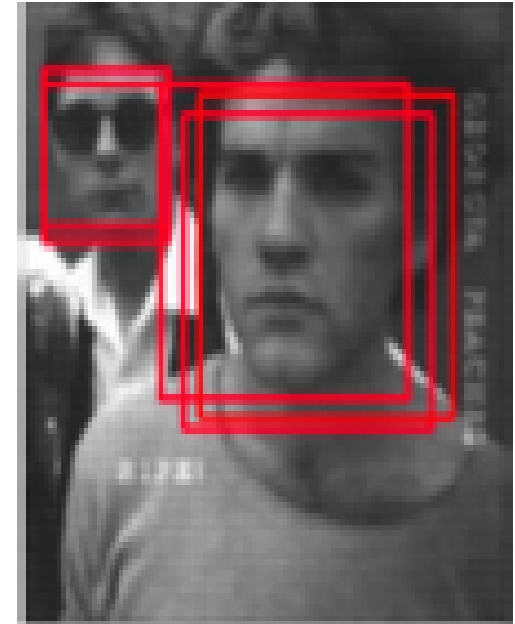
Training Data
of Non-Face

Face Detection Results



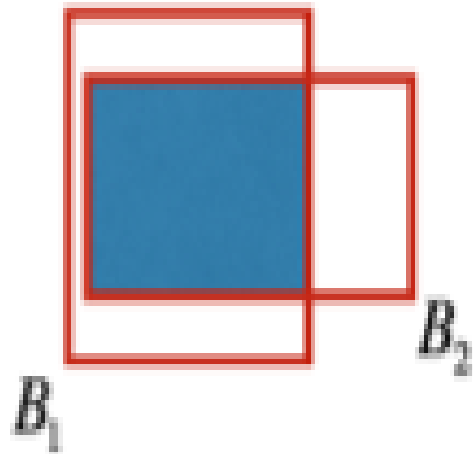
Non-Maximum Suppression

- Usually, more than one bounding box is detected for same object
- Use bounding box similarity measures to identify duplicate bounding boxes which can be removed

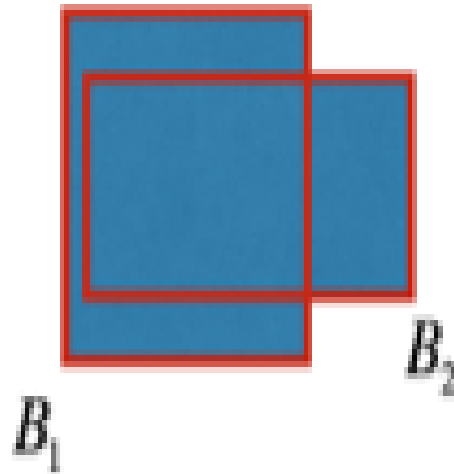


Intersection over Union (IoU)

Intersection



Union



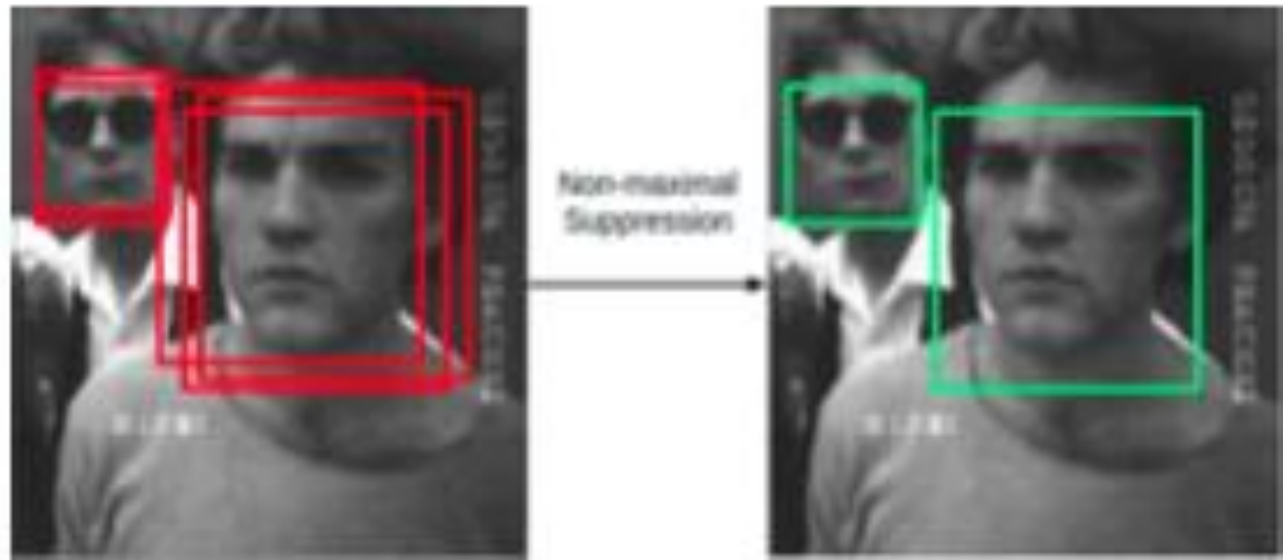
Intersection over Union

$$IoU = \frac{B_1 \cap B_2}{B_1 \cup B_2} = \frac{\text{Intersection}}{\text{Union}}$$

A diagram illustrating the components of the IoU formula. The numerator is represented by a small blue square (the intersection) inside a red-outlined square. The denominator is represented by two overlapping blue squares (the union) inside red-outlined squares. A horizontal line separates the numerator and denominator.

Non-Maximum Suppression

- Select a random box from a bounding box proposal list
- Compare it with rest of the boxes; if $\text{IoU} > 0.5$, remove box from list
- Repeat until boxes left are exclusive



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- Thanks to the following researchers for making their teaching/research material online
 - Forsyth
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 - R. Zaleski
 - Leibe
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